Co-ordinated Multi Agent Reinforcement Learning for Autonomous Driving in Mixed Traffic Environments

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by

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Abstract

Autonomous driving has opened the door to a new era of transportation, but it has also brought new challenges. In today's roads, self-driving cars rarely operate in isolation; instead, they share space with human drivers, each with their own unpredictable behaviours. This thesis explores a promising approach to help autonomous vehicles navigate this mixed-traffic reality: **Coordinated Multi-Agent Reinforcement Learning (CMARL)**. By training vehicles to not only do things independently, but to do things in coordination with others, CMARL provides a means of addressing the complex combination of instantaneous decision-making and collaboration that these environments require. The goal is to create a decentralized, scalable, and coordination-friendly model for autonomous driving using CMARL, across various traffic scenarios such as mergers, stoplight intersections, roundabouts, and platooning.

The methodology employs centralized training with decentralized execution (CTDE) paradigm utilizing a modified Multi-Agent Deep Deterministic Policy Gradient (MADDPG++) algorithm, alongside rule-based evaluation, scenario-specific modular training, and graph-based coordination. This research uses customized environments simulated in a lightweight physics engine and is extensible to environments such as CARLA or SUMO. A variety of human-like and rule-based agents are introduced to model real-world dynamics to test for adaptability. The evaluation will consider multiple measures of coordination, collision rate, reward trends, and generalization across unseen environments through zero-shot testing. The most significant findings demonstrate that coordination-aware training greatly enhances the safety, convergence speed, and adaptability of autonomous agents in mixed-autonomy scenarios. This modular architecture also allows for the easier incorporation of rules, facilitates zero-shot evaluation, aids explainability, and makes it suitable for potential future use in real-time simulations or hardware-in-the-loop evaluations.

This thesis contributes a scalable and extensible architecture for CMARL in autonomous driving, a novel training-evaluation pipeline with scenario-aware abstraction, and empirical results validating coordination benefits in complex traffic scenarios. The research underscores the potential of coordinated learning strategies to enable safer and more intelligent autonomy in real-world transportation systems.

Keywords: Multi-Agent Reinforcement Learning (MARL), Autonomous Driving, Mixed-Traffic Coordination, MADDPG++, Scenario-Based Simulation, Rule-Based Evaluation.

Foreword

The journey toward this master's thesis has been both a professional pursuit and a personal voyage one shaped by curiosity, perseverance, and an enduring fascination with intelligent systems. As I think about the journey that brought me to this point, I'm struck by the number of times, moments, questions, and experiences that have helped me develop my knowledge about artificial intelligence and the everyday world we navigate.

The idea of vehicles that perceive, decide, and adapt on their own was once confined to the realm of science fiction. Yet as I am immersed into the study of machine learning and autonomous systems, this idea turned into a material technical problem, requiring both ingenuity and accountability. I felt inspired by the prospect that machines can learn to coordinate, act with limited information, and operate in safety interdependently with humans. It has been the idea of efficient, intelligent mobility in mixed-autonomy traffic that has motivated this work.

Coming from a background where technology was often viewed through a utilitarian lens, I found immense inspiration in the elegance of multi-agent systems. Their complexity, coupled with their potential for scalable coordination, mirrored not just traffic dynamics but the broader patterns of collective intelligence in nature and society. This thesis, in many ways, became my attempt to decode those patterns and bring them to life through code, architecture, and learning algorithms.

Working on Coordinated Multi-Agent Reinforcement Learning (CMARL) for autonomous driving has been more than an academic task it has been an invitation to contribute, in however small a way, to a future of safer and smarter mobility. It is a field that demands not only technical rigor but also ethical reflection and interdisciplinary understanding, values that I strived to embody throughout this work.

I would like to extend my heartfelt gratitude to Prof. Dr. Alexander Iliev, whose mentorship, vision, and belief in my potential gave me the confidence to pursue this ambitious topic. I am equally indebted to my peers, collaborators, and academic community for their insightful discussions, constructive criticism, and unwavering encouragement.

To my family and close friends, thank you for your patience, your faith, and your constant support during moments of doubt and discovery alike. This thesis would not have come to life without your presence behind the scenes.

As I now hand over this document to you, the reader, I hope you will find within its pages not just algorithms and results, but a sincere exploration of cooperation between intelligent agents, one that aspires, in its own small way, to advance our shared journey toward a more connected and autonomous future.

Venkat Chavan
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Special appreciation is due to the open-source and AI research communities whose contributions made it possible for independent researchers like me to experiment, learn, and innovate. I acknowledge the role of advanced AI tools such as ChatGPT, Gemini, and Perplexity, which served as intelligent collaborators during moments of ideation, code refactoring, literature synthesis, and clarity-seeking. These tools, though virtual, offered me the sense of not being entirely alone in the intellectual journey.

To my family, especially my parents, thank you for your unconditional belief in me and for standing by my side through every challenge. To my friends who listened patiently, encouraged me, and reminded me to breathe, your support meant more than words can express.

This thesis is dedicated to all self-driven learners who build alone fueled by passion, persistence, and the quiet belief that even solitary efforts can shape meaningful futures.

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Abbreviations

Acronym	Explanation
ADAS	Advanced Driver Assistance Systems
AI	Artificial Intelligence
AV	Autonomous Vehicle
BC	Behavioral Cloning
CARLA	Car Learning to Act (autonomous driving simulator)
CMARL	Coordinated Multi-Agent Reinforcement Learning
CTDE	Centralized Training with Decentralized Execution
CV	Computer Vision
DDPG	Deep Deterministic Policy Gradient
DRL	Deep Reinforcement Learning
FL-RL	Federated Reinforcement Learning
GAT	Graph Attention Network
GATConv	Graph Attention Convolution
GNN	Graph Neural Network
GPU	Graphics Processing Unit
HDV	Human-Driven Vehicle
HITL	Human-in-the-Loop
LiDAR	Light Detection and Ranging
LLM	Large Language Model
MADDPG	Multi-Agent Deep Deterministic Policy Gradient
MADDPG++	Enhanced MADDPG with Graph-based extensions
MAPPO	Multi-Agent Proximal Policy Optimization
MARL	Multi-Agent Reinforcement Learning
MDP	Markov Decision Process
ML	Machine Learning
MLP	Multi-Layer Perceptron

Acronym	Explanation
MORL	Multi-Objective Reinforcement Learning
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
RSU	Roadside Unit
SAE	Society of Automotive Engineers
Sim2Real	Simulation to Reality Transfer
SLAM	Simultaneous Localization and Mapping
SUMO	Simulation of Urban Mobility
V2I	Vehicle-to-Infrastructure
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything
VLM	Vision-Language Model

1. Introduction

1.1. Background and Subject Relevance

1.1.1. Introduction to Autonomous Driving

Autonomous driving, once a speculative domain of science fiction, has evolved into a frontier of cutting-edge technological innovation. Enabled by advances in artificial intelligence (AI), sensor technologies, connectivity, and computational power, autonomous vehicles (AVs) are envisioned to revolutionize modern transportation systems. These vehicles can perceive their environment, make decisions, and execute control actions with minimal or no human intervention (Litman, 2020). The societal implications are profound, ranging from enhanced road safety and increased accessibility to reduced congestion and improved energy efficiency.

Figure 1 - Structure of an autonomous driving system showing perception, planning, control, and actuation layers (IEEE, 2024).

The transformation from human-driven to autonomous mobility is occurring in incremental stages, defined by the Society of Automotive Engineers (SAE) from Level 0 (no automation) to Level 5 (full automation) (SAE International, 2018). While current commercial applications are predominantly at Level 2 or 3, the research community continues to advance capabilities that will support fully autonomous driving systems in complex real-world

environments. The growing number of pilot deployments by companies like Waymo, Tesla, and Baidu underscores both the feasibility and the challenge of scaling AV technology (Darapaneni, 2021).

1.1.2. Core Components of Autonomous Driving

At a high level, autonomous driving systems integrate four major functions: perception, prediction, planning, and control. Perception involves interpreting data from sensors such as LiDAR, radar, cameras, and GPS to create a representation of the surrounding environment. Prediction refers to anticipating the future actions of other road users. Planning computes the trajectory or path the vehicle should take, and control executes this plan through steering, acceleration, and braking (Paden, 2016).

These modules must operate in real time, be robust to uncertainties, and comply with traffic rules. The difficulty of the problem is magnified in dynamic, multi-agent scenarios where interactions with other vehicles, especially human-driven ones, introduce non-determinism and behavioral variability.

1.1.3. The Role of Reinforcement Learning in AVs

Conventional autonomous driving systems have historically been built upon rule-based frameworks and carefully engineered heuristics. While such approaches can be highly effective in structured or predictable settings, their adaptability diminishes when faced with unfamiliar or unpredictable scenarios. This limitation has prompted increasing interest in Reinforcement Learning (RL), a branch of machine learning that enables agents to develop optimal behaviors through iterative interaction with their environment (Sutton & Barto, 2018).

In the RL paradigm, an agent repeatedly engages with its environment by observing its current state, selecting an action, and receiving feedback in the form of a reward that reflects the quality of that action. Over successive interactions, the agent refines its decision-making strategy, or *policy*, with the objective of maximizing the cumulative reward over time. Deep Reinforcement Learning (DRL) advances this framework by incorporating neural networks to approximate complex value functions and policies, thereby enabling the handling of large-scale, high-dimensional state and action spaces that are common in autonomous driving contexts.

1.1.4. From Single Agent to Multi Agent Reinforcement Learning

While single-agent reinforcement learning can effectively address problems in controlled, isolated settings, autonomous vehicles function within highly interactive, shared environments. In such contexts, decision-making must account not only for the agent's own state but also for the likely intentions, actions, and responses of other participants. Multi-Agent Reinforcement Learning (MARL) expands the traditional RL framework to accommodate scenarios in which multiple agents learn and adapt simultaneously (Buşoniu, Babuška, De Schutter, & Ernst, 2008).

$$\boldsymbol{J(\theta) = E\pi\theta\left[\sum_{t=0}^{T} \gamma^t r_t\right]}$$

Equation 1: Reinforcement Learning Objective.

In MARL, each agent aims to optimize its policy while engaging with other agents in a continuously evolving environment. This interdependence introduces several well-recognized challenges:

- **Non-stationarity:** Because other agents' strategies change throughout the training process, the environment appears non-stationary from the perspective of any single agent.
- **Scalability:** The joint state–action space increases exponentially with the number of agents, making computation and learning increasingly complex.
- **Credit assignment:** In cooperative settings, it can be difficult to determine the specific contribution of each agent to the overall system's performance.

1.1.5. Centralized Training with Decentralized Execution (CTDE)

A widely adopted design principle in Multi-Agent Reinforcement Learning is Centralized Training with Decentralized Execution (CTDE), which facilitates scalable and efficient coordination among agents (Lowe, et al., 2017). Under this framework, training is conducted with access to global information allowing each agent to consider the observations and actions of all others. In contrast, execution is performed in a fully decentralized manner, where each agent makes decisions based solely on his own local observations. This design reflects the practical constraints of real-world traffic systems, in which vehicles cannot access complete global state information during operation.

$$\nabla_\theta J(\theta) = E_{\pi\theta}[\nabla_\theta log\pi_\theta(a \mid s)Q^\pi(s,a)]$$

Equation 2 - Policy Gradient.

CTDE has served as the foundation for several influential MARL algorithms, including:

- **MADDPG** – Multi-Agent Deep Deterministic Policy Gradient
- **QMIX** – A value-based method for cooperative multi-agent tasks
- **MAPPO** – Multi-Agent Proximal Policy Optimization

$$Qi\mu(s, a_1, \ldots, a_N) = E[ri + \gamma Qi\mu'(s', a'_1, \ldots, a'_N)]$$

Equation 3 - Multi-Agent Deep Deterministic Policy Gradient (MADDPG) Critic

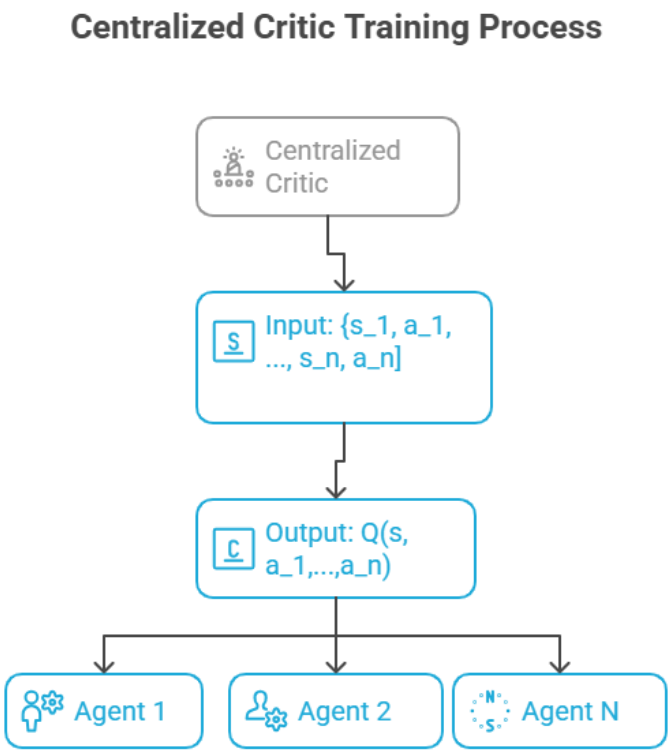

Figure 2 - Illustration of the Centralized Training with Decentralized Execution (CTDE) paradigm, where training leverages global information but execution remains decentralized (adapted from (Lowe, et al., 2017); (Zhou, et al., 2022)).

1.1.6. Coordination in Mixed-Traffic Scenarios

Mixed-traffic environments where autonomous and human-driven vehicles share the roadway pose some of the most significant challenges for autonomous vehicle deployment. Human drivers often display a wide range of behaviours, many of which are unpredictable and deviate from strictly rational or model-based assumptions. In such conditions, effective

coordination is essential not only to minimize the risk of collisions but also to reduce decision-making delays and enhance overall traffic flow efficiency.

1.1.7. Graph Neural Networks in MARL for AVs

Given that traffic systems inherently possess relational and network-like structures, Graph Neural Networks (GNNs) have become a well-suited approach for modelling these environments (Li, Chen, Xu, & Zhou, 2020). By capturing the underlying dependencies and interactions among agents, GNNs can improve situational awareness and adaptive decision-making capabilities that are particularly valuable in high-interaction scenarios such as roundabouts, lane merges, or freeway on-ramps.

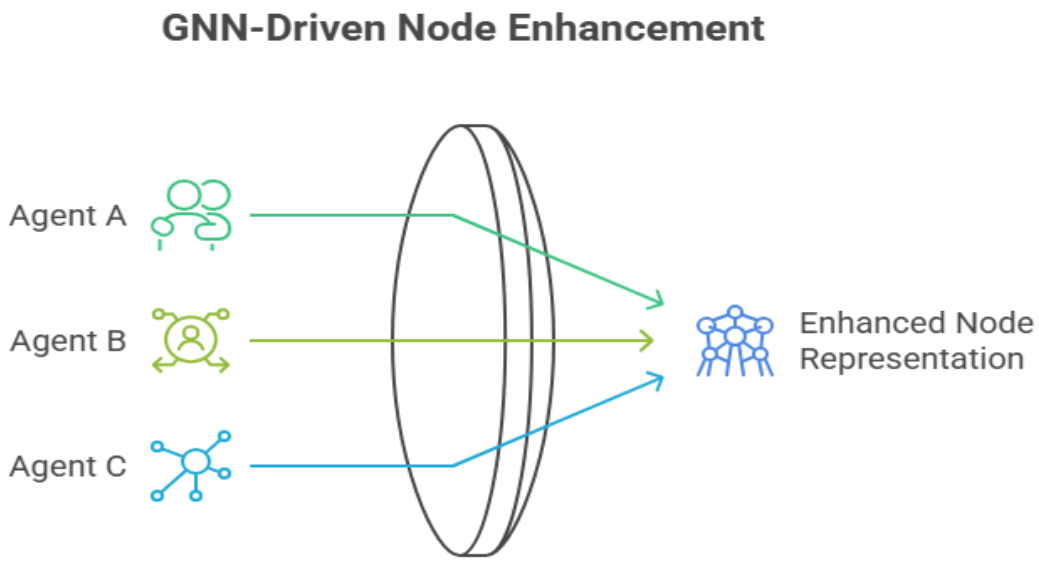

Figure 3 - Message passing in Graph Neural Networks (GNNs), where agents exchange information through edges to generate enhanced node representations (adapted from (Li, Chen, Xu, & Zhou, 2020)).

1.1.8. Rule-Based vs Learning-Based Decision Making

In autonomous vehicle decision-making, existing approaches can be broadly divided into two categories:

- Rule-based systems, which rely on predefined logic and explicitly coded behavioral rules.
- Learning-based systems, which adapt their strategies through data-driven methods and experience.

Recent research increasingly supports hybrid frameworks that combine the strengths of both paradigms using learning-based methods to optimize decision-making while incorporating rule-based modules to ensure adherence to safety constraints and regulatory requirements.

1.1.9. Simulation Platforms and Benchmarks

Because autonomous driving is inherently safety-critical, most MARL research takes place in simulation rather than on real roads. Simulation platforms like CARLA, SUMO, and Highway-env provide customizable virtual environments where researchers can safely test and refine autonomous vehicle behaviour across a wide range of traffic scenarios.

1.1.10. Practical Relevance and Societal Impact

Bringing Coordinated MARL systems into real-world autonomous driving pipelines could unlock significant societal gains making roads safer, traffic more efficient, and systems more scalable and adaptable to changing conditions.

1.2. Motivation

My personal motivation for this research stems from an enduring fascination with intelligent systems that can not only learn but *collaborate*. As a student of Big Data and AI, I've always been intrigued by the challenge of engineering machine coordination in uncertain, human-dominated environments. Growing up in traffic-dense regions where road unpredictability is the norm, I often wondered how autonomous systems could ever adapt to such chaos. This curiosity matured into a commitment to explore AI architectures capable of real-time cooperation and safe decision-making. Through this thesis, I aim to contribute meaningfully to the ongoing discourse on multi-agent intelligence in safety-critical systems.

1.3. Research Gap

Despite notable advancements in Multi-Agent Reinforcement Learning (MARL) for autonomous driving, several critical limitations persist that impede the real-world deployment of coordinated decision-making systems in mixed-traffic environments. Existing MARL frameworks such as MADDPG (Lowe, et al., 2017), QMIX, and MA2C have demonstrated promising results in cooperative or homogeneous agent settings, particularly in simulation-based benchmarks like CARLA and SUMO. However, their assumptions and constraints limit their generalizability to more realistic, heterogeneous traffic conditions.

1. Limited consideration of heterogeneous and non-cooperative agents:

Most MARL studies focus on collaborative agents that follow shared policies or communication protocols. However, real-world roads involve non-cooperative human drivers, whose behaviors are inherently stochastic, irrational, and context dependent. Very few models simulate such mixed-autonomy interactions accurately. As pointed out in your proposal, coordination strategies tend to break down in the presence of human drivers who neither share policies nor provide reliable communication feedback.

2. Sim-to-real transferability remains unsolved:

While high-fidelity simulators like CARLA offer realism, many MARL algorithms trained in simulation suffer from poor transferability to the physical world. This "reality gap" arises from over-reliance on idealized assumptions, such as perfect sensing, full observability, and rule-based agents. (Liu, 2022) emphasize that even with advanced simulation, personality traits of human drivers are hard to model accurately, leading to reduced effectiveness in safety-critical tasks.

3. Underexplored coordination under partial observability and V2X limitations:

Real-world agents often operate under limited sensing ranges, sensor occlusions, or communication failures. Existing MARL implementations rarely address implicit coordination with the ability to learn cooperative behaviors without explicit message passing. (Zhou, et al., 2022)noted that although multi-agent actor-critic methods can address these to some extent, they typically require extensive parameter tuning and fail under low-information scenarios.

4. Lack of generalizable frameworks across diverse traffic topologies:

Most existing models are scenario-specific designed for intersections, merges, or highway exits in isolation. This restricts their application in cities with varied traffic geometries and driving cultures. Your proposal explicitly calls attention to the need for a single CMARL framework that can generalize across roundabouts, intersections, and highways while adapting to local agent density, vehicle types, and behavior profiles.

5. Safety and ethical alignment remain an open challenge:

Reinforcement learning methods often optimize for rewards like speed or efficiency, which can lead to unsafe emergent behaviors (e.g., tailgating, cutting-in) unless explicitly constrained. Few MARL implementations apply constrained MDPs (CMDPs) or ethical priorities. (Valiente, 2022) highlights that robustness under uncertainty and noise, especially in mixed-autonomy settings, remains insufficiently explored.

6. Absence of cross-cultural adaptability:

Driving behaviors and traffic norms vary significantly across regions. However, most existing MARL studies train and test models in uniform, culturally agnostic scenarios, ignoring factors like aggression levels, gap acceptance, or right-of-way behavior. Without accounting for these, MARL agents may underperform or behave inappropriately in unfamiliar environments.

1.4. Objectives and Goals

The primary objective of this thesis is to investigate and develop a robust multi-agent reinforcement learning system that enables coordinated behavior among autonomous vehicles in mixed-traffic environments. The goals are:

- To implement and evaluate CTDE-based MARL algorithms such as MADDPG and its graph-enhanced variants for traffic coordination.
- To model realistic traffic scenarios (lane change, merge, intersection, roundabout, platooning) that demand cooperation and conflict resolution.
- To simulate human-like agents alongside autonomous ones for mixed-autonomy evaluation.
- To design an ASD rule-based evaluation module for tracking violations, rewards, and safety scores.
- To test zero-shot generalization capabilities by training in one scenario and deploying in another.
- To analyze emergent behaviors, cooperation metrics, and convergence properties under varied traffic densities.

1.5. Research Questions

The central research questions this thesis seeks to answer are:

1. How can reinforcement learning be used to enable coordinated decision-making among autonomous vehicles in complex traffic environments?
2. What are the comparative strengths and limitations of CTDE-based algorithms like MADDPG in multi-agent driving tasks?
3. How can traffic-rule compliance and safety-critical behavior be evaluated within a learning-based multi-agent framework?
4. To what extent do trained agents generalize about novel driving environments without retraining?
5. Can rule-based agents and learning-based agents interact safely and effectively in shared road scenarios?

1.6. Scope & Limitations

This thesis focuses on cooperative MARL strategies for decision-making in simulated traffic environments with up to 6 agents (autonomous and human-like). It leverages discrete-time episodic learning on simplified road geometries such as merges, intersections, and roundabouts using a Python-based simulation platform.

In scope:

- Development of modular MARL training pipelines.
- Design of discrete multi-agent traffic scenarios.
- CTDE learning strategies (MADDPG and GNN-based variants).
- Evaluation with custom rule-based metrics.
- Simulation-based validation (no real-world deployment).

Out of scope:

- Physical deployment on real vehicles or hardware-in-the-loop testing.
- Sensor-level perception and control (e.g., camera-LiDAR fusion).
- Full vehicle dynamics modeling (e.g., tire models, suspension).

- Adversarial training or defensive driving under attack conditions.

2. Literature review & theoretical framework

2.1. Literature review

The intersection of autonomous driving (AD) and reinforcement learning (RL) has evolved over the past two decades, shaped by foundational advances in robotics, perception, and intelligent transportation systems. Early milestones, such as the DARPA Grand Challenge in the 2000s, laid the foundation for autonomous navigation in unstructured environments (Thrun, 2006). The subsequent rise of deep learning (LeCun, 2015), coupled with the proliferation of scalable computing resources, catalyzed the integration of RL into dynamic and complex driving tasks.

By the mid-2010s, deep reinforcement learning (DRL) algorithms like Deep Deterministic Policy Gradient (DDPG; (Lillicrap, et al., 2016)) and Asynchronous Advantage Actor-Critic (A3C; (Mnih V. B., 2016)) enabled the learning of high-dimensional control policies in simulated environments. Building on these foundations, more sophisticated methods such as Multi-Agent Deep Deterministic Policy Gradient (MADDPG; (Lowe, et al., 2017)) were developed to address coordination challenges in multi-agent settings. These advances increasingly aligned with the demands of connected autonomous vehicles (CAVs), where agents must operate cooperatively under uncertainty in heterogeneous traffic scenarios.

2.2. Literature Themes and Synthesis

The reviewed literature can be categorized into five interrelated domains:

- Multi-Agent Systems in Autonomous Driving
- Reinforcement Learning Applications for Traffic Control
- Coordination in Mixed-Traffic Scenarios
- Theoretical Algorithmic Foundations (MDP, MARL, CTDE)
- Application Extensions: Autonomous Racing, Blockchain Security, Edge Deployment

Each domain is discussed below with critical references.

2.2.1. Multi Agent Systems in Autonomous Driving

Multi-agent systems (MAS) in autonomous driving empower individual vehicles modeled as autonomous agents to learn and execute policies that may be cooperative, competitive, or hybrid in nature. Recent research, such as that by (Wang J. H., 2022) and (Zhu, 2021), highlights the effectiveness of decentralized policy execution, especially in complex environments like intersections, where centralized control can be impractical or computationally burdensome.

Notably, the (Vehicle control using cooperative decision making, 2020) patent outlines a cooperative vehicle framework based on agent-level decision-making modules and a shared representation of the environment an approach that aligns with the paradigm of centralized training with decentralized execution (CTDE). This design enables vehicles to learn coordinated strategies during training while acting autonomously in real time.

A core strength of MAS lies in its capacity to model emergent behavior, wherein agents autonomously learn to perform maneuvers such as lane merging, overtaking, or right-of-way negotiation without relying on hardcoded, rule-based instructions. Such adaptive behaviors are essential for ensuring safety, scalability, and responsiveness in mixed-traffic environments.

2.2.2. Reinforcement Learning Applications in Traffic and Driving

Reinforcement learning (RL) has emerged as a powerful framework for addressing the complexities of dynamic driving environments. Algorithms such as Deep Deterministic Policy Gradient (DDPG), Twin Delayed DDPG (TD3), and Proximal Policy Optimization (PPO) have been widely applied to continuous control tasks, with reward functions tailored to promote safety, efficiency, and passenger comfort (Sallab, 2017(19)).

Recent works have emphasized the importance of reward shaping in guiding agents to adopt socially compliant and context-aware driving behaviors, such as safe lane changes and smooth path planning. In parallel, multi-agent extensions of Q-learning (Lauer, 2000) have shown promise in enabling cooperative strategies for tasks like lane merging and multi-lane highway coordination.

2.2.3. Coordination Strategies in Mixed Traffic Environments

Coordination in mixed-autonomy traffic where autonomous vehicles (AVs) and human-driven vehicles coexist presents unique challenges due to partial observability, behavioral noise, and decision-making uncertainty. (Liniger, 2021) examined such dynamics in high-speed racing simulations, where emergent behaviors like aggressive overtaking and defensive blocking highlighted the spectrum of learned driving strategies in competitive environments.

To address the need for trust and accountability in multi-agent scenarios, (Jain, et al., 2021) proposed integrating blockchain technology into AV systems, enabling transparent, tamper-proof decision logging and real-time negotiation. This is particularly relevant in applications such as ride-sharing coordination, platooning, and intersection management involving multiple stakeholders.

Moreover, graph neural networks (GNNs) are increasingly being adopted to model inter-agent dependencies, especially in constrained settings like intersections or merge lanes. By encoding relational and spatial structures, GNNs enable agents to learn more context-aware and cooperative policies (Wang J. C., 2020), enhancing safety and adaptability in dense traffic environments.

2.2.4. Algorithmic Foundations

Foundational frameworks such as Markov Decision Processes (MDPs) and Partially Observable Markov Decision Processes (POMDPs) provide a mathematical basis for modeling sequential decision-making under uncertainty and stochastic transitions. In multi-agent settings, these models extend to Decentralized POMDPs (Dec-POMDPs), which account for the distributed nature of observations and actions across agents, while also introducing significant coordination and scalability challenges (Oliehoek & Amato, 2016).

To address these challenges, the paradigm of Centralized Training with Decentralized Execution (CTDE) has emerged as a widely adopted standard in multi-agent reinforcement learning (MARL). CTDE allows agents to leverage global information and shared gradients during training, while maintaining decentralized policies during execution making it particularly suitable for real-world applications like the scenario explored in this thesis.

2.2.5. Related Applications: Racing, Blockchain and Edge Development

Autonomous racing has emerged as a high-speed benchmark for algorithmic performance. As noted in (Betz, et al., 2022), racing agents must operate near physical limits of acceleration and latency, making them ideal for testing RL generalization and adversarial planning.

Edge computing, as discussed by (Jain, et al., 2021), enables low-latency policy execution and distributed critical logging for scenarios where bandwidth or cloud access is limited.

Blockchain, while peripheral, provides secure consensus and multi-agent auditability, especially in use-cases involving vehicle-to-vehicle contracts or cooperative maneuver validation.

2.2.6. Major Contributions and Synthesis Table

Research Area	**Key Contributions**	**Algorithms**	**Relevance**
AV Perception	HD-mapping, cone detection, SLAM in Formula Student (Betz, et al., 2022)	GraphSLAM, YOLOv3	Environment mapping
Planning & Control	Minimum-time racing line, local obstacle avoidance	MPC, RRT*	Coordination at high speeds
MARL	Shared policy learning, CTDE-based MADDPG	MADDPG, MAPPO	Core to CMARL
Blockchain	Distributed AV consensus	Pow, Smart Contracts	Audit & security layer
Edge Learning	Low-latency inference	FL + DNN	Decentralized deployments

Table 1: Major Contributions and Synthesis

2.2.7. Year-by-Year Publication Trends

Year	Key Themes
2015	DDPG, Deep RL for continuous control
2017	MADDPG introduced, Dec-POMDPs gain traction
2019	RL in autonomous vehicles, MARL simulations using SUMO
2020	CTDE formalized, advances in actor-critic architectures
2021	Autonomous racing, integration of blockchain in AV decision-making
2022	GNNs for MARL, Federated MARL, Human-in-the-loop autonomous driving
2023	Hybrid rule-based + RL coordination, Coordinated MARL (CMARL) applications
2024	Emergent strategies in mixed autonomy, GAT-enhanced MADDPG models

Table 2 - Trends in Research

2.3. In-Depth Research Areas Relevant to CMARL

2.3.1. Deep Dive into Multi Agent reinforcement Learning in Autonomous Driving

The application of Multi-Agent Reinforcement Learning (MARL) to autonomous driving has become a cornerstone of contemporary research in traffic coordination. Autonomous driving is inherently a multi-agent problem, as multiple vehicles must simultaneously perceive, plan, and act within a shared environment. In such settings, an autonomous vehicle must not only ensure its own safety and efficiency but also adapt to the interactive behaviors of surrounding vehicles, pedestrians, and infrastructure making MARL a naturally fitting paradigm.

As noted by (Zhou, et al., 2022), MARL enables agents (vehicles) to develop policies that are robust to environmental uncertainties and capable of adapting to interactions with both cooperative and non-cooperative entities. In contrast to single-agent reinforcement learning, which assumes a stationary environment, MARL acknowledges that the environment evolves as other agents learn and act giving rise to the non-stationarity problem, wherein the policy updates of one agent can interfere with the learning processes of others.

To mitigate this challenge, algorithms such as Multi-Agent Deep Deterministic Policy Gradient (MADDPG; (Lowe, et al., 2017)) introduced the use of centralized critics during training, allowing for more stable convergence while maintaining decentralized policies for execution. This structure laid the foundation for the now-dominant Centralized Training with Decentralized Execution (CTDE) paradigm, which has been further extended by approaches such as Multi-Agent Proximal Policy Optimization (MAPPO) and Multi-Agent Actor-Critic (MAAC).

2.3.2. Taxonomy of Muti Agent Reinforcement Learning Approaches

Imagine a busy intersection where both autonomous and human-driven cars are navigating simultaneously. To understand how self-driving vehicles learn to cooperate and make decisions in such complex environments, we can categorize their learning systems along several dimensions:

Training Architecture

- **Centralized Training:** Think of this as one coach training the entire team using a single playbook. All vehicles learn from the same shared experiences and rules.
  Pros: Efficient knowledge sharing ensures consistent behavior across cars.
  Cons: As the number of vehicles grows, managing and scaling this centralized approach becomes challenging like coaching a large sports team where individualized attention is limited.
- **Decentralized Training:** Here, each vehicle has its own instructor and learns independently based on its unique experiences on the road.
  Pros: Allows vehicles to adapt to their specific contexts.
  Cons: Different training paths can make coordination difficult, akin to players each following their own game plan without a shared strategy.
- **Centralized Training with Decentralized Execution (CTDE):** This approach is like holding a team meeting to discuss strategies together, but once the game starts, each player makes decisions independently based on local observations.
  Pros: Combines shared learning benefits with autonomous execution, balancing coordination and flexibility.

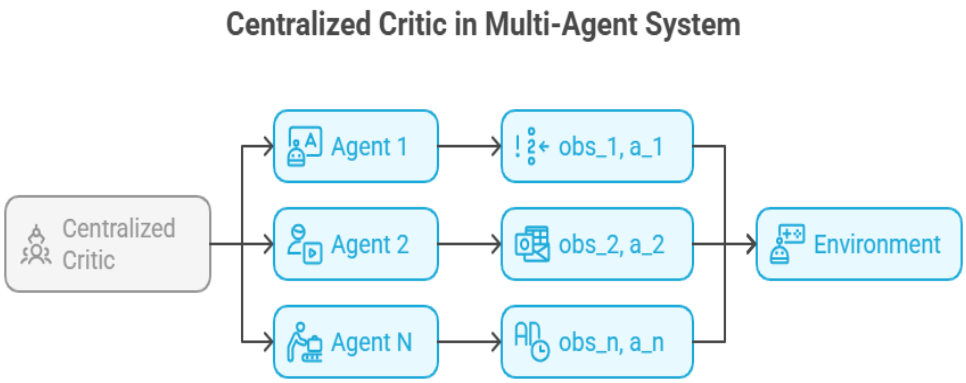

Figure 4 - Centralized Training with Decentralized Execution (CTDE) framework, where agents train using a centralized critic, but act based on local observations (adapted (Lowe, et al., 2017)).

Communication Type

- **Implicit Communication:** Vehicles interpret subtle cues from each other such as noticing a slowing car and inferring a potential turn.
  Pros: Enables natural, fluid interactions without explicit messaging.
  Cons: Requires highly sensitive perception and inference capabilities, which can be challenging to implement reliably.
- **Explicit Communication:** Vehicles directly share intentions like broadcasting" I am changing lines"
  Pros: Makes coordination clear and reduces uncertainty.
  Cons: Depends on reliable, low-latency communication infrastructure.

Reward Structure

- **Shared Reward Structure:** All the vehicles work towards common goals such as maximizing overall safety and traffic like earning a team bonus for group success
  Pros: Encourage cooperation and collective benefit
  Cons: May overlook individual vehicle objective constraints
- **Individual Rewards:** Each vehicle receives rewards based on its own performance, encouraging, personalized goals like safe driving or energy efficiency.
  Pros: Useful in mixed environments with diverse vehicles objectives.
  Cons: Can foster competition that may undermine cooperation

Agent Homogeneity

- **Homogeneous Agents:** All vehicles are similarly designed and trained, following a united framework like identical players on a team.
  Pros: Simplifies training and system management
  Cons: Increase complexity of coordination and integration
- **Heterogeneous Agents:** Vehicles differ in design, sensors, and learning methods like a diverse team where each member contributes unique skills.
  Pros: Offers flexibility and tailored solutions
  Cons: Increase complexity of coordination and integration

By classifying multi-agent autonomous driving systems along these dimensions, researchers and engineers can better design cooperative frameworks that enable safe, efficient, and adaptive traffic management helping to build smarter roads and smoother commutes.

2.3.3. Coordination Strategies in Mixed Traffic Environments

In real-world traffic, the coexistence of human-driven and autonomous vehicles introduces significant challenges for coordination under uncertainty. Recent studies, including (Zhou, et al., 2022), have highlighted the complexity arising from interactions between rule-based human-like agents and learning-based autonomous agents. Their findings demonstrate that such hybrid interactions produce emergent patterns that traditional control frameworks struggle to capture accurately.

For instance, in intersection scenarios, autonomous vehicles (AVs) may learn to yield to aggressive human drivers while adopting a more assertive policy when the surrounding environment is predominantly cooperative. This adaptive behavior emerges naturally only when agents are trained on a diverse distribution of heterogeneous driving behaviors.

A promising direction involves constructing personality models of human drivers through behavior cloning techniques, which are then used to populate training environments for multi-agent reinforcement learning (MARL) agents, as proposed by (Fang, Wu, & Sun, 2024). This approach enhances the ability of AVs to generalize across varying human behavioral patterns.

Additionally, explicit coordination is facilitated by vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication protocols. Systems like Autonomous Intersection

Management (AIM) leverage real-time negotiation of right-of-way by sharing state estimates and broadcasting intent, enabling efficient and safe distributed traffic management.

2.3.4. Graph Neural Networks and Interaction-Aware Learning

In environments where agents must reason about spatial and temporal dependencies, Graph Neural Networks (GNNs) have emerged as a powerful modeling framework. In this approach, each agent is represented as a node in a graph, while edges encode spatial proximity, communication links, or interaction relevance between agents.

(Wang J. H., 2022) demonstrated that employing attention-based GNN architectures, such as Graph Attention Convolution (GATConv) layers, enables agents to dynamically prioritize information from their most influential neighbors a mechanism analogous to social attention. This capability is particularly advantageous in dense urban settings, where it is crucial for agents to discern which surrounding vehicles are most relevant for immediate decision-making.

GNN-based policies also offer strong scalability: they naturally generalize to environments with varying numbers of agents and changing traffic densities. Such adaptability is essential for the real-world deployment of multi-agent autonomous traffic systems, where coordination must remain robust across diverse and dynamic operational contexts.

2.3.5. Reinforcement Leaning for Cooperative Driving Tasks

Several cooperative driving tasks have been implemented using MARL:

- Merging onto highways: Requires implicit negotiation of gap sizes, yielding behavior.
- Roundabout Navigation: Needs joint timing estimation to avoid collisions.
- Platooning: Requires precise longitudinal control with low-latency communication.

In these cases, MARL agents are trained with objectives such as:

$$R_t = \alpha_1 \cdot velocity_t + \alpha_2 \cdot gap_safety_t - \alpha_3 \cdot collision_t$$

Equation 4 - Training Objectives in MARL

Where:

- $\alpha_1, \alpha_2, \alpha_3$ are reward shaping weights
- $velocity_t$ encourages speed efficiency
- gap_safety_t rewards safe following distances
- $collision_t$ penalizes any collision

These reward functions are essential for shaping behaviours that mimic human-like driving patterns while optimizing for safety and flow.

2.3.6. Use of Simulators in Literature

A wide variety of simulators have been employed in MARL driving research:

- **SUMO**: Lightweight, scalable, great for lane/route planning.
- **CARLA**: High-fidelity physics and rendering, ideal for testing vision and sensor fusion.
- **Flow**: Bridges SUMO and RLlib, allowing training of RL agents in realistic traffic.
- **Petting Zoo**: A general-purpose MARL environment with discrete support.

In the studies discussed above, they respectively used these simulators to prototype and benchmark agent performance while reviewing collisions and evaluate how coordination would emerge.

2.3.7. Limitations of Existing Approaches

Despite the advancements made in Multi-Agent Reinforcement Learning (MARL) and its application in autonomous driving, there are still technical, theoretical, and practical constraints that ultimately limit its scalability, safety, and generalization in mixed-traffic environments in real-world application.

Non-Stationarity and Convergence Instability

A central challenge in MARL is the non-stationarity of the environment from the perspective of each individual agent. As all agents simultaneously update their policies during training, the environment's dynamics continuously shift, making stable learning difficult. Algorithms such as MADDPG (Lowe, et al., 2017) and QMIX (Rashid, et al., 2018) address this issue through centralized critics, which provide a global perspective during training. However,

convergence instability remains a persistent issue particularly in high-dimensional observation spaces or under sparse reward signals.

Moreover, MARL methods demand careful tuning of hyperparameters such as learning rates, exploration-exploitation trade-offs, and reward shaping to ensure stable convergence. Poor tuning can lead to divergence or convergence to locally optimal yet globally suboptimal policies. Additionally, the multi-agent credit assignment problem identifying which agent's actions contributed to a collective outcome further complicates training, especially in cooperative tasks where rewards are shared or delayed.

Poor Transferability and Generalization

A critical barrier to real-world deployment of MARL-based autonomous vehicle (AV) systems is their limited generalization across diverse driving scenarios and agent configurations. Policies trained in a specific simulation environment such as SUMO often exhibit poor transferability to others like CARLA or real-world datasets, due to overfitting to environment-specific dynamics, visual representations, or assumptions (Chen C. Y., 2021).

In reality, there is a lot of variability between concrete driving environments. From differences in weather conditions to road infrastructure, rules of the road, and even regional driving behaviors, all can significantly change how a policy performs. As an example, a model trained to navigate roundabouts in Germany may not perform in the same fashion when facing four-way stops in the United States and relies on right-of-way logic. This lack of robustness represents an anthropological safety risk to safety and scalability in heterogeneous real-world traffic ecosystems.

Scalability Constraints

As the number of agents in a traffic environment increases, coordination complexity and computational demands grow rapidly particularly when models rely on centralized critics or joint Q-functions. Techniques such as parameter sharing and graph-based representations (e.g., GATConv) have made meaningful progress in reducing this overhead. However, the scalability limits of current MARL approaches remain a pressing issue, especially when applied to large-scale traffic systems involving dense and dynamic agent interactions (Wang J. H., 2022).

Notably, most of the MARL research is still confined to experimental setups involving fewer than 20 agents. In contrast, real-world traffic scenarios routinely involve hundreds or even

thousands of vehicles, where interactions are not only local but also long-range and influenced by delayed feedback. Bridging this gap between simulation-scale and real-world complexity is essential for practical deployment.

Limited Modelling of Human Behaviour and Intent

Many MARL approaches oversimplify mixed-traffic scenarios by treating other vehicles either as fellow reinforcement learning agents or as static, rule-based obstacles. In reality, human drivers make up most road users, and their behavior is anything but predictable shaped by personal habits, emotional states, and social context.

While recent efforts using imitation learning or inverse reinforcement learning (IRL) have aimed to replicate human driving patterns, they often fall short of capturing deeper latent intent or non-verbal social signals like eye contact, assertive maneuvering, or courteous yielding (Fang, Wu, & Sun, 2024). Without explicitly modeling such nuanced human behaviors and bounded rationality, MARL-based policies risk producing unintuitive or even unsafe decisions in real-world interactions.

Reward Misalignment and Shaping Bias

Reward shaping plays a crucial role in defining the emergent behavior of agents. However, hand-crafted rewards can inject designer bias and often fail to align with global safety or fairness goals.

Consider the following sample reward function used in lane change scenarios:

$$r_t = w_1 \cdot efficiency_t - w_2 \cdot collision_t + w_3 \cdot comfort_t$$

Equation 5 - Reward Signal.

While this may work in simulations, it lacks adaptability to contexts like emergency braking or erratic human maneuvers. Misalignment between reward signal and real-world utility remains an open challenge.

Lack of Formal Guarantees and Safety Verification

Unlike traditional rule-based systems, MARL methods pose significant challenges when it comes to formal verification and validation. This is especially problematic for autonomous driving, where even rare failures can result in catastrophic consequences. Ensuring that learned policies behave safely and reliably across edge cases, adversarial conditions, or distributional shifts is a critical yet unresolved issue.

Although techniques such as adversarial training, shielding mechanisms, and runtime verification have shown promise, they are often applied in isolation. A comprehensive and holistic safety framework for MARL in real-world traffic environments remains largely absent from the current research landscape (Jahan, 2021) (Pokhrel, 2021).

Deployment and the Simulation-to-Reality (Sim2Real) Gap

While high-fidelity simulators provide a controlled and scalable environment for training MARL agents, bridging the gap between simulation and real-world deployment remains a core challenge. Real-world settings introduce a host of complexities such as sensor noise, communication latency, and actuation delays that are rarely captured in simulation. As a result, policies that perform well in environments like CARLA or SUMO often fail to generalize when deployed on actual vehicles.

Moreover, edge deployment brings its own set of constraints: limited computation and bandwidth, partial observability, and asynchronous decision-making. Current MARL frameworks frequently overlook real-time feedback loops between agents and infrastructure (e.g., roadside units, traffic signals), further widening the Sim2Real gap. Overcoming these limitations demands advanced domain adaptation techniques, robust transfer learning strategies, and tighter integration between simulation models and real-world sensor-actuator pipelines.

Summary of Identified Limitations

Limitation	**Underlying Cause**
Non-Stationarity	Dynamic policy shifts during training
Scalability Bottlenecks	Centralized critics and joint action spaces
Human-Agent Discrepancy	Lack of intent modeling and social cues
Reward Misalignment	Over-reliance on static shaping terms
Poor Generalization	Overfitting to narrow environment setups
Simulation-to-Reality Gap	Sensor noise, latency, and domain drift

Limitation	Underlying Cause
Safety Verification	Opaque policies lacking formal guarantees

Table 3 - Core Limitations in Multi-Agent Reinforcement Learning (MARL)

These limitations justify the need for modular, robust, and explainable MARL pipelines, such as the one proposed in this thesis. Designs intended for future living must consider integration of adaptive reward models, realistic human-agent interactions, and cross-environment evaluation for safe and scalable deployment.

2.3.8. Emerging Trends and Research Opportunities in MARL for Autonomous Driving

As self-driving technology is advancing rapidly, the progression of Multi-Agent Reinforcement Learning (MARL) in mixed-traffic applications is quickly maturing. Recent advancements include innovative new coordination designs and hybrid learning processes, and substantial theoretical policy performance guarantees. In overcoming previously highlighted shortcomings, the emerging research trends are now paving the way for MARL in the next generation of MARL solutions, for real-world applications to autonomous vehicle systems.

Human-in-the-Loop (HITL) and Behavioural Cloning Integration

A significant advancement in Multi-Agent Reinforcement Learning (MARL) is the integration of human feedback into the training process. Human-in-the-Loop (HITL) reinforcement learning enables agents to refine their policies using human demonstrations, critiques, or preference-based feedback, rather than relying solely on reward signals from the environment. This approach fosters more interpretable and socially aligned behaviors in complex, real-world scenarios.

For instance, (Jeon, 2022) propose a framework that combines inverse reinforcement learning (IRL) with policy shaping, allowing agents to approximate human preferences in intricate driving tasks such as lane merging or yielding. This hybrid strategy lays the foundation for aligning agent behavior with human social driving norms, which is essential for safe and cooperative autonomy.

Moreover, behavioral cloning (BC) offers a data-driven method to bootstrap MARL agents using large-scale, real-world driving datasets such as the INTERACTION dataset or the

Waymo Open Dataset. By initializing policies through supervised learning on expert trajectories, BC significantly reduces the risk of unsafe exploration in early training phases and encourages the emergence of socially plausible behaviors before transitioning to reinforcement-based fine-tuning.

Transfer Learning and Domain Adaptation for Sim2Real

Bridging the gap between simulation-trained policies and real-world deployment remains a key challenge in Multi-Agent Reinforcement Learning (MARL). Recent research has increasingly focused on transfer learning and domain adaptation techniques to address this sim-to-real (Sim2Real) transfer problem.

One prominent approach is domain randomization, where simulation parameters such as lighting conditions, road surface properties, and sensor noise are varied during training. This helps agents generalize better to unseen real-world variations. Another promising direction is meta-learning, which equips agents with the capability to quickly adapt to new environments or tasks with minimal retraining. Additionally, teacher-student frameworks have gained traction where a teacher agent trained extensively in simulation transfers knowledge to a student agent deployed in the real world, often through distillation or guidance mechanisms.

For instance, (Zhao X. S., 2023) introduce a domain-adaptive policy distillation method that fine-tunes policies trained in simulation (e.g., using CARLA) onto real-world platforms. Their approach leverages differential sensor modalities between simulation and physical environments to facilitate effective policy adaptation, demonstrating improved robustness and safety in real-world testbeds.

Graph-Based and Attention-Based Architectures

Recent advances in Multi-Agent Reinforcement Learning (MARL) have increasingly embraced Graph Neural Networks (GNNs) especially those enhanced with attention mechanisms such as GATConv to better model spatial relationships and inter-agent dynamics in traffic environments.

In approaches like G-MADDPG and MA-GAT, agents construct dynamic interaction graphs based on criteria such as proximity or influence. These graphs help each agent selectively attend to its most relevant neighbors, improving coordination in dense, complex environments. As a result, agents can exhibit emergent behaviors like adaptive yielding,

convoy formation, or context-aware overtaking driven by their understanding of the local traffic topology.

Importantly, these graph-based methods naturally support variable numbers of agents and offer greater scalability across different scenarios. This flexibility addresses a longstanding limitation in traditional flat MARL architectures, where rigid communication patterns or fixed agent structures often impede generalization to real-world traffic systems.

Decentralized Federated MARL

As area of connected infrastructure keeps expanding, there has been greater interest in federated reinforcement learning (FL-RL), which allows decentralized policy learning in multiple agents like autonomous vehicles (AVs) and roadside units (RSUs) while ensuring data privacy and independence from the system.

Rather than relying on a central server, FL-RL frameworks allow each agent to train locally on its own data and periodically share model updates with others. This setup significantly reduces the communication burden and mitigates privacy concerns associated with raw data sharing. For instance, (Pokhrel, 2021) propose blockchain-integrated federated learning architecture, allowing AVs to securely exchange policy updates without centralized aggregation. Their framework fosters trustworthy coordination among multiple parties in shared driving environments.

This decentralized model is particularly appropriate for the emerging Vehicle-to-Everything (V2X) protocols, where AVs can use their own experience and learn from the overall experiences of surrounding agents in real-time that leads to the safer, smarter and more cooperative decision making on the road.

Multi-Objective and Safe MARL

In real-world autonomous driving, agents are rarely tasked with a single goal. Instead, they must balance multiple often competing objectives, such as minimizing travel time, maximizing fuel efficiency, ensuring passenger comfort, and maintaining safety. To address this complexity, Multi-Objective Reinforcement Learning (MORL) has emerged as a promising direction within the broader MARL landscape.

Rather than optimizing a single scalar reward, MORL models operate on a vector-valued reward function, where each component represents a different objective.

$$\overrightarrow{R_t} = [r_{safety}, r_{comfort}, r_{efficiency}, r_{adherence}]$$

Equation 6 - Vector Valued Reward Function

This setup allows agents to consider trade-offs between objectives using principles like Pareto optimality and constrained policy optimization, ensuring that no one goal is improved at the severe expense of another.

To further enhance safety, some frameworks incorporate RL shields or safety layer mechanisms that monitor and override potentially dangerous actions during training or deployment, without halting learning altogether (Ray, 2019). These safeguards are particularly valuable in high-stakes scenarios like autonomous driving, where even a single unsafe maneuver could have serious consequences.

Realistic Driver Behaviour and Emotion Modelling

A huge challenge for self-driving cars is making sure they can handle real-world drivers, who often act illogically. After all, people aren't always calm and rational behind the wheel. The next big step for autonomous vehicles (AVs) is to model these unpredictable, emotional behaviors.

New research is tackling this by using a few different methods. Some approaches use probabilistic behavior trees or Bayesian intent prediction to forecast what a driver might do. Others are even using deep personality modeling networks (PeMN) to capture the full range of human unpredictability.

By training AVs on these realistic driver behaviors everything from road rage to distracted driving to sudden braking their policies become much more robust and safer. A paper by (Valiente, 2022) highlights this, arguing that AVs need to understand how human drivers interact with them to be truly safe and cooperative on the road.

Reward Learning from Observations

Training self-driving cars to make good decisions is hard because it's nearly impossible to write down a perfect set of rules, or a "reward function," for every situation they might face. A much more promising approach is rewarding learning.

Instead of a programmer defining the reward function, the car learns it by watching how a human driver handles different situations. It figures out the underlying goal that best explains

human actions. This is done through techniques like maximum entropy Inverse Reinforcement Learning (IRL) or Bayesian reward inference.

This method is useful for several driving challenges, such as:

- **Handling intersections**: How to safely and efficiently get through a busy junction.
- **Lane changes**: Knowing when and how to merge smoothly.
- **Balancing speed and comfort**: Finding the sweet spot between getting to the destination quickly and giving passengers a comfortable ride.

In the future, this process could be improved even further by asking people to rate or compare different driving scenarios, which would provide even more guidance for the car's learning.

Explainability and Trust in MARL Agents

Self-driving cars use complex AI and understanding how they make decisions is a big problem. These systems are often called "black boxes" because their logic isn't transparent. To build trust and meet safety standards, we need to be able to explain their actions.

This is where explainability comes in. Tools like SHAP (SHapley Additive exPlanations) and graph attention visualization help us peer inside the black box. They allow us to see which factors the AI considered most important when deciding. This is critical for:

- **Debugging**: If an autonomous vehicle (AV) does something wrong, we can use these tools to figure out why, making it easier to fix.
- **Trust**: Passengers, safety testers, and government regulators can see the reason behind AV’s actions. This transparency is key to building confidence in technology.

The proposal to use graph attention visualization fits right into this trend. It's a method that helps explain the AI's decision-making process, which is exactly what future regulations will require for autonomous vehicles.

Hierarchical and Curriculum-Based Learning

Learning to drive is hard enough, and for an AI, it's even more complicated, especially in a busy city. Instead of trying to teach an AI, everything at once, a promising approach is hierarchical and curriculum-based learning.

Think of it like a student driver. You wouldn't throw them into downtown traffic on day one. You'd start with simple tasks in a parking lot and gradually move to more complex situations. This is curriculum learning.

Similarly, hierarchical reinforcement learning (HRL) breaks down the driving task into two main parts:

1. **A high-level planner** acts like the brain, deciding on the bigger picture, like "I need to get to the next intersection" or "I should find an open lane to merge."
2. **A low-level controller** acts like the hands and feet, figuring out the small, immediate actions to achieve that bigger goal, such as "turn the steering wheel a few degrees" or "accelerate slightly."

By combining these two ideas, AI drivers can learn complex skills more effectively. This has shown great results in challenging virtual environments like City Flow and the CARLA Challenge, which simulate real urban traffic.

Interdisciplinary Research and Digital Twins

A rapidly emerging trend in the MARL landscape is the integration of digital twin platforms, which create real-time, high-fidelity virtual replicas of physical traffic environments by synchronizing with live sensor data from autonomous vehicles and infrastructure.

This fusion of simulation and reality enables continuous policy learning and real-world validation. By testing MARL agents in digital twins, researchers can safely evaluate and refine driving behaviors under realistic operational constraints bridging the gap between controlled experiments and chaotic, dynamic roadways.

Beyond simulation, this trend signals a broader interdisciplinary convergence. The integration of IoT, edge AI, intelligent transportation systems (ITS), and secure communication protocols (e.g., blockchain) highlights the need for collaboration across domains including artificial intelligence, civil and traffic engineering, robotics, cybersecurity, and public policy.

As MARL systems become part of large-scale, interconnected urban ecosystems, their success increasingly depends not just on algorithmic breakthroughs but also on holistic, cross-domain innovation and governance.

Emerging Trends in MARL for Autonomous Driving

TREND	IMPACT
Human-in-the-loop RL	Preference safety
Transfer learning / Sim2Real	Deployment-ready
Graph-based MARL	Scalable attention
Federated MARL	Privacy & trust
Multi-objective optimization	Balanced driving
Human modeling (PeMN)	Realistic behavior
Reward inference	Adaptive learning
Explainability tools	Trust + debugging
HRL + Curriculum	Task hierarchy
Digital Twins	Real-world sync

Table 4 - Summary Table of the Trends

2.4. Theoretical Framework

2.4.1. Foundations of Decision Making in Autonomous Agents

Markov Decision Process (MDPs) in Autonomous Driving

An autonomous agent's decision-making process can be modelled using a Markov Decision Process (MDP), defined formally as the tuple:

$$\boldsymbol{M = \langle S, A, P, R, \gamma \rangle}$$

Equation 7 - MDPs

Where:

- S is the set of all possible states (e.g., position, speed, proximity to other vehicles)
- A is the set of available actions (e.g., accelerate, decelerate, change lanes)
- $P: S \times A \times S \rightarrow [0,1]$ is the transition probability function

- $\mathcal{R}: S \times A \times S \rightarrow \mathbb{R}$ are the reward functions
- $\gamma \in [0,1]$ is the discount factor for future rewards

This formulation assumes full observability, which is often unrealistic in dynamic traffic conditions.

Partially Observable MDPs (POMDPs)

To better reflect real-world driving where agents have limited perception (e.g., occlusions, sensor noise), the MDP is extended to a POMDP:

$$\mathbf{P} = \langle S, A, P, R, \Omega, O, \gamma \rangle$$

Equation 8 - POMDPs

- Ω is the set of observations and
- $O : S \times A \rightarrow \Omega$ maps states and actions to observations

This abstraction supports agent modelling in limited field-of-view and sensor-imperfect simulations (Zhou, et al., 2022).

2.4.2. Multi-Agent Reinforcement Learning (MARL)

Multi-agent systems for autonomous vehicles require coordination strategies that scale with agent count and heterogeneity. MARL enables learning behaviours through interaction among agents, where each agent is typic (Iqbal, 2019) ally modelled as an independent learner but may coordinate via shared rewards or communication (Iqbal & Sha, 2019).

2.4.3. Centralized Training with Decentralized Execution (CTDE)

Formalism

CTDE facilitates **coordinated learning** by allowing agents to share global state during training but act independently at execution time:

$$\boldsymbol{Q_i(s, a_1, \ldots, a_N)\ during\ training\ \rightarrow\ \pi_i(o_i)\ during\ execution}$$

Equation 9 - Execution time of CTDEs

Here:

- Q_i is the critic that access the joint-state action space
- $\pi_i(o_i)$ is the decentralized policy based on agent i's local observation of o_i

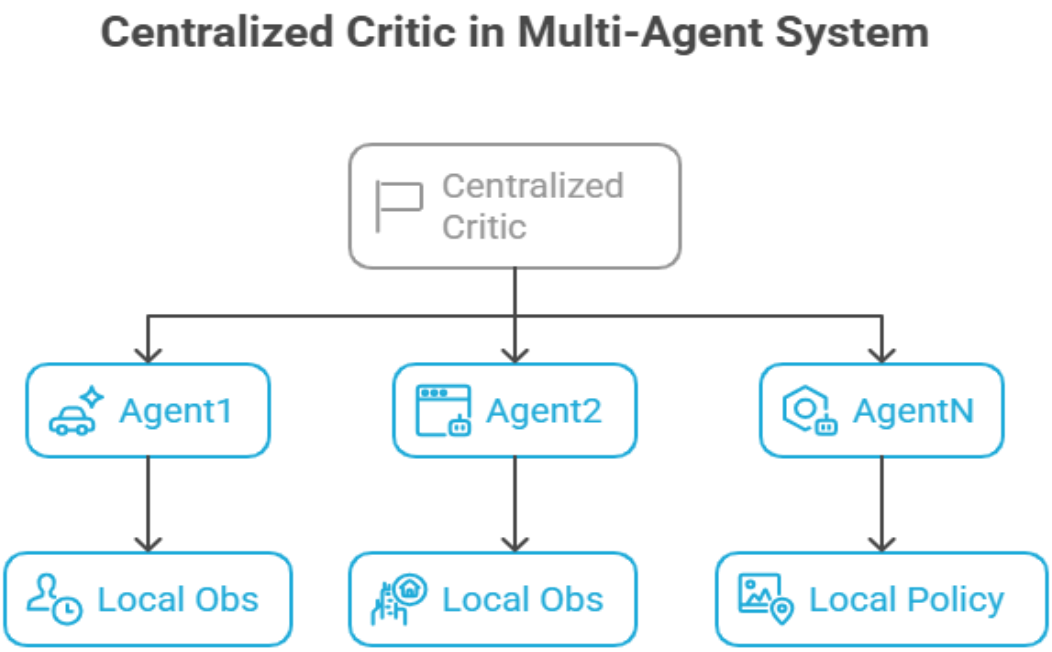

Figure 5 - Centralized training with decentralized execution (CTDE), where the centralized critic leverages joint information while agents execute based on local observations (adapted from (Lowe, et al., 2017); (Foerster, 2018)).

This paradigm supports scalability in heterogeneous and mixed-autonomy environments, aligning well with the thesis goal.

2.4.4. Deep Deterministic Policy Gradient (DDPG)

Core Equations

DDPG (Lillicrap et al., 2016) is an actor-critic, off-policy algorithm for continuous action spaces.

- Critic Update: $L(\theta^Q) = \mathbb{E}_{s_t,a_t,r_t,s_{t+1}}[(Q(s_t, a_t|\theta^Q) - y_t)^2]$

Equation 10 - Critic Update

Where: $y_t = r_t + \gamma Q'\left(\left(s_{t+1}\,|\theta^{\mu'}\right) \mid \theta^{Q^1}\right)$

- Actor Gradient: $\nabla_{\theta_\mu} J \approx \mathbb{E}_{s_t}\left[\nabla_a Q(s, a|\theta^Q)|_{a=\mu(s)} \cdot \nabla_{\theta^\mu}\mu(s|\theta^\mu)\right]$

Equation 11 - Actor Gradient

2.4.5. Multi Agent DDPG(MADDPG)

MADDPG extends DDPG to multi-agent settings by giving critical access to the joint action and joint state: $Q_i(s, a_1, \ldots\ldots, a_N)$

Each actor $\mu_i(o_i)$ only takes local observations. MADDPG has been foundational in recent work on mixed-traffic navigation, particularly under CTDE schemes (Wang J. H., 2022).

2.4.6. Multi Agent Proximal Policy Optimization (MAPPO)

MAPPO builds on the popular PPO algorithm by extending it to multi-agent environments with a shared policy architecture (Yu et al., 2021). Its objective:

$$L^{\boldsymbol{CLIP}}(\theta) = \mathbb{E}_t \left[min\left(r_t(\theta)\hat{A}_t, clip(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t\right)\right]$$

Equation 12 - MAPPO

Where:

- $r_t(\theta)$ is the probability ratio,
- $\hat{A}_t$is the advantage estimate
- ϵ is the hyperparameter (typically 0.1-0.3)

MAPPO improves training stability and sample efficiency, especially in dense reward environments (Xu, Gao, Yu, Darrell, & Malik, 2020).

2.4.7. Reward Shaping in Mixed Traffic Environment

Effective reward design is crucial for convergence and safety. Some examples from the literature include:

$$r_t = \alpha r_{self} + \beta r_{coop}, \qquad where\ \alpha + \beta = 1$$

$$r_{self} = f_{lane_keeping} + f_{speed} + f_{collision_avoidance}$$

$$\boldsymbol{r_{coop}} = \sum_{\boldsymbol{j \neq i}} \boldsymbol{success_of(j)}$$

Equation 13 - Rewards

This approach allows encoding individual rationality and team objectives simultaneously ((Zhou, et al., 2022); (Toghi, 2021)).

2.4.8. Communication and Coordination via GNNs

Graph Neural Networks (GNNs) enhance agent interaction modelling by capturing structured relationships. Each agent is represented as a node, and edges encode interaction (e.g., proximity or intent)

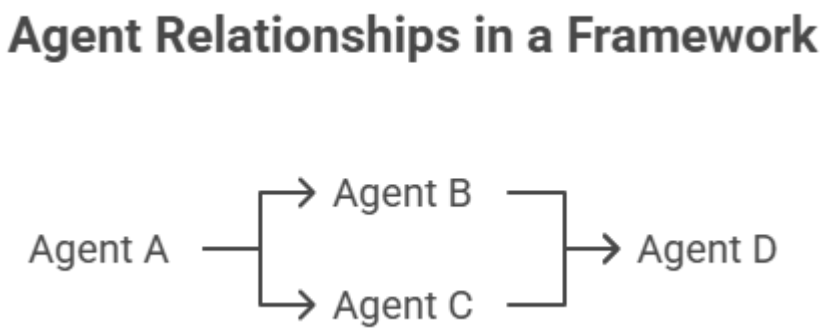

Figure 6 - Network of agent interactions represented as a graph, where nodes denote agents (A–D) and edges encode communication or coordination dependencies (own illustration, based on the GNN coordination concept from (Veličković, 2018)).

Each node aggregates messages from its neighbors using attention (Veličković, 2018):

$$h_i^{(l+1)} = \sigma\left(\sum_{j \in \mathrm{N}(i)} \alpha_{ij} W h_j^{(l)}\right)$$

Equation 14 - GATConv

Where:

- $h_j^{(l)}$ is the node representation at layer l
- α_{ij} are attention weights computed via SoftMax over similarity scores

In your GATConv-based MADDPG++, this formulation enables agents to learn whom to attend to dynamically, enhancing robustness under uncertainty.

3. Methodology

3.1. Overview of the Methodological Approach

This research develops a Coordinated Multi-Agent Reinforcement Learning (CMARL) framework aimed at improving autonomous driving in mixed-traffic environments. The central objective is to enable vehicles to make safe, cooperative decisions while interacting with a variety of agent types, including both autonomous and human-driven vehicles.

The framework follows a Centralized Training with Decentralized Execution (CTDE) paradigm. During training, agents benefit from access to global information, allowing them to learn cooperative strategies. At deployment, however, each vehicle acts independently based only on its local observations, ensuring scalability and autonomy.

At the algorithmic level, the study builds on an enhanced version of the Multi-Agent Deep Deterministic Policy Gradient (MADDPG++) method. This is extended with graph attention layers, which enable structured communication between agents. By using attention-based message passing, vehicles can assign more importance to those neighbors that are contextually relevant, for example, a nearby car closing in at high speed while ignoring less critical agents. This design ensures that coordination adapts dynamically to evolving traffic situations.

The methodology is structured around six scenario-specific simulation environments, each reflecting a distinct class of traffic interaction:

1. **Merge Environment** - highway merging with lane discipline and right-of-way compliance
2. **Intersection Environment -** unprotected intersections with multi-directional flows
3. **Lane Change Environment -** lateral movement under varying speeds and gaps
4. **On-Ramp Merge Environment-acceleration Lane** merging into high-speed flows
5. **Platooning Environment -** cooperative speed harmonization and gap control
6. **Roundabout Environment -** multi-entry circular flow negotiation

Each environment was developed as an independent module with its own state space, action definitions, and reward design. A custom Gym-style API was used to ensure modularity and compatibility with the training pipeline. This modular setup also enabled zero-shot

generalization tests, where models trained in one scenario were evaluated in others to assess adaptability. The training process is organized into episodes comprising the following sequence:

- **State Acquisition** - Each agent perceives its local environment via an observation vector, including kinematic variables, relative distances, and rule-based flags for ASD compliance
- **Action Selection** - Decentralized actors generate continuous control commands, augmented by Gaussian noise for exploration
- **Environment Update** - Actions are applied, and the environment simulates the resulting dynamics
- **Reward Computation** - Rewards are assigned based on safety, efficiency, and cooperation, shaped by ASD-compliant traffic rules
- **Experience Storage** - Transitions are stored in a replay buffer for off-policy learning
- **Policy & Value Updates** -The centralized critic updates policies using batch samples, applying gradient clipping and soft target updates to stabilize learning

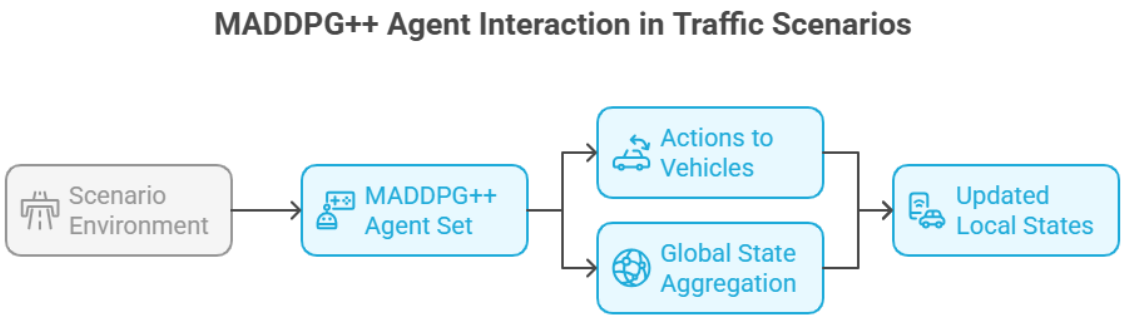

Figure 7 - High-level CMARL workflow under CTDE, showing centralized critic during training and decentralized execution during deployment

This methodology is reinforced by three critical extensions to enhance robustness and interpretability:

- **Human-in-the-Loop Integration** - selected agents operate under human-defined or rule-based logic to simulate realistic mixed-autonomy traffic
- **Explainability via Graph Attention Weights** - Visualization of attention coefficients provides interpretability into coordination strategies and decision-making logic

- **ASD Rule-Based Evaluation Module** - Post-training evaluation enforces compliance with traffic rules and quantifies violations, bridging the gap between pure reward optimization and real-world safety standards

In essence, this approach merges deep reinforcement learning, graph-based coordination, and rule-constrained evaluation into a single pipeline, producing agents capable of both competitive performance and regulatory compliance in complex driving environments.

3.2. System Architecture

The proposed Coordinated Multi-Agent Reinforcement Learning (CMARL) system is implemented as a modular and scenario-independent architecture. It is designed to flexibly support different types of agents, a wide range of traffic situations, and varying levels of coordination.

At the core lies an enhanced Multi-Agent Deep Deterministic Policy Gradient (MADDPG++) framework, extended with graph attention layers that enable structured communication between agents. These layers allow the system to capture relational dependencies in traffic such as which vehicles are most relevant for immediate decision-making while filtering out less critical interactions.

The overall design follows the principle of Centralized Training with Decentralized Execution (CTDE). In this setup, agents are trained with access to global state information, which allows them to learn cooperative and safety-aware behaviors. Once deployed, however, each agent acts autonomously based on its local observations, ensuring scalability and independence in real-world traffic.

3.2.1. Overall CMARL Architecture

The system comprises the following high-level components (Figure 7):

1. Scenario Environment Module

Each driving scenario (Merge, Intersection, Lane Change, On-Ramp Merge, Platooning, Roundabout) is implemented as a self-contained environment following an OpenAI Gym-compatible API. This enables uniform interaction with the training loop, where each environment:

- Generates an initial state with randomized traffic conditions
- Advances the simulation in time steps based on agent actions
- Compute rewards from safety, efficiency, and cooperation metrics
- Signals episode termination based on scenario completion or violation

2. Agent Module

Each autonomous agent contains:

- Actor Network - Maps observed states to continuous control actions (e.g., acceleration, steering)
- Critic Network - Estimates the Q-value for the agent's actions, conditioned on both local observations and the aggregated global state during training
- GATConv Layer - Processes graph-structured agent interactions, applying attention mechanisms to dynamically weigh neighboring agents' influence (Veličković, 2018)

3. Communication Graph Construction

At each step, the positions and velocities of agents are used to construct an adjacency matrix defining which agents are considered "neighbors" for message passing. The GATConv layer computes:

$$h_i' = \sigma\left(\sum_{j \epsilon N(i)} \alpha_{ij} W h_j\right)$$

Equation 15 - GATConv Layer

Where α_{ij} are learned attention coefficients, W is a shared weight matrix and σ is a nonlinear activation function

4. Replay Buffer

A shared experience replay buffer stores tuples o_t, a_t, r_t, o_{t+1} for all agents, enabling off-policy learning and sample reuse.

5. Centralized Critic Decentralized Actor

During training, the critic receives concatenated observations and actions from all agents, providing richer context for gradient updates:

$$Q_i(o_1, \ldots\ldots, o_N, a_1, \ldots\ldots, a_N; \theta_i^Q)$$

Equation 16 - Centralized Critic Decentralized Actor

During execution, each actor operates solely on its local observation o_i.

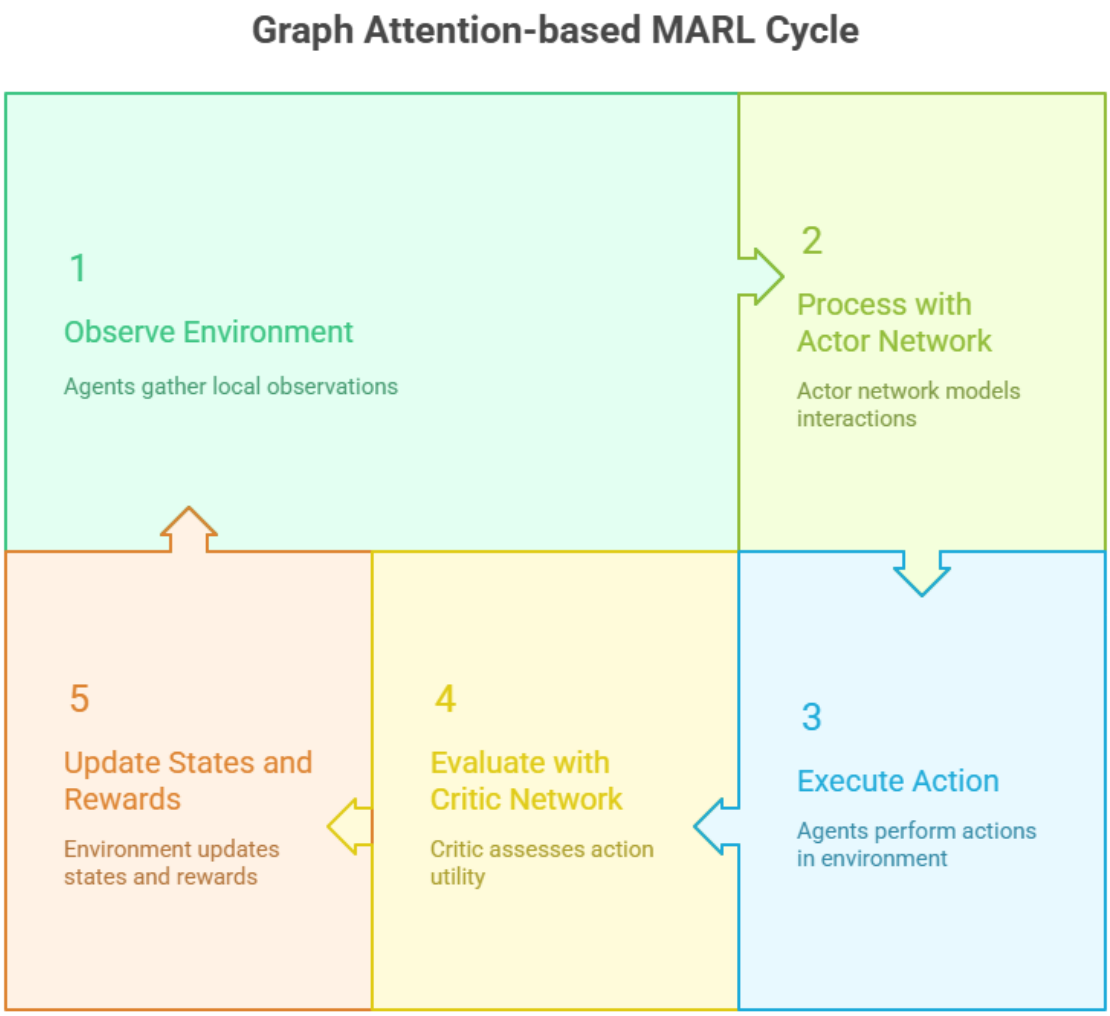

Figure 8 - High-level CMARL system architecture with graph attention-based message passing, illustrating the iterative cycle of observation, actor processing, action execution, critic evaluation, and state update

3.2.2. Centralized Training with Decentralized Execution (CTDE)

A central challenge in multi-agent reinforcement learning (MARL) is non-stationarity: as each agent updates its policy during training, the effective environment changes for all other agents. This often leads to instability and poor convergence if not addressed.

The CTDE paradigm provides a practical solution. During training, a centralized critic is used so that each agent's value function can condition on the full joint state–action space.

This stabilizes learning by allowing agents to reason with complete information about the system.

At deployment, however, agents must act in real-world traffic without access to global states. To ensure scalability and autonomy, the execution phase is fully decentralized: each actor relies solely on its local observations to make decisions. This balance between centralized learning and decentralized operation makes CTDE especially suited for large-scale, mixed-traffic environments.

Actor Update Rule

The deterministic policy gradient for agent i is:

$$\nabla_{\theta_i^\mu} J(\mu_i) \approx \mathbb{E}_{o,a\sim D}\left[\nabla_{\theta_i^\mu}\ \mu_i(o_i)\nabla_{a_i} Q_i(o, a_1, \ldots\ldots, a_N)|_{a_i=\mu_i(o_i)}\right]$$

Equation 17 - CTDE Actor Update Rule

Critic Update Rule

The critic minimizes the temporal-difference (TD) error:

$$L(\theta_i^Q) = \mathbb{E}_{o,a,r,o'}[(Q_i(o,a) - y_i)^2]$$

Equation 18 - CTDE Critic Update Rule

With: $y_i = r_i + \gamma Q_i'(o', a_1', \ldots\ldots, a_N')$

Where Q_i' and a_1' are target, networks updated via: $\theta' \leftarrow \tau\theta + (1-\tau)\theta'$ for smoothing factor $\tau \ll 1$.

3.2.3. GATConv Based Communication

The GATConv layer enables each agent to:

- Aggregate neighbor information (e.g., speed, lane, proximity).
- Assign learned attention weights αij\alpha_{ij}αij to focus on the most influential agents.
- Dynamically adapt communication structure across steps.

This is especially beneficial in mixed-traffic conditions, where relevant agents change rapidly (e.g., a merging vehicle becomes more critical than a distant platoon member). In

your code, the adjacency matrix is recomputed each timestep, ensuring temporal adaptability in communication.

3.3. Environment and Scenario Design

The CMARL framework is deployed across six custom driving scenarios, each modeled as an independent environment module following the OpenAI Gym interface specification. This modular approach enables targeted training, comparative benchmarking, and zero-shot generalization across diverse traffic contexts. All environments integrate Autonomous System Design (ASD)-compliant rules to ensure safety and regulatory adherence during both training and evaluation.

3.3.1. Merge Environment

Objective

To simulate highway merging behaviour, where agents must blend into a main lane while maintaining safety distances and optimizing traffic flow efficiency.

State Space:

- Ego vehicle kinematics (position, velocity, acceleration)
- Relative positions and speeds of neighboring vehicles in merging zone
- Lane availability flags (binary indicators)
- Rule compliance flags (e.g., safe gap availability)

Action Space:

- Continuous acceleration $[-1,1]$
- Steering adjustments $[-0.5,0.5]$ radians

Reward Design:

- **Positive:** Successful merge without collision, smooth speed adaptation
- **Negative:** Collision, forced braking of other vehicles, unsafe merges

Special Features:

- To increase realism in the simulation environments, two additional features were implemented. First, human-driven vehicles (HDVs) are introduced with randomized

spawning and varying levels of aggressiveness. This variability creates diverse interaction patterns and forces autonomous agents to adapt their strategies to different driving styles. Second, the framework incorporates dynamic gap creation during merging situations. When multiple autonomous vehicles are present, they learn to coordinate by adjusting speed and spacing to create safe entry gaps, reflecting cooperative behaviour required in real-world traffic. Intersection Environment

Objective

To coordinate multi-directional traffic at unprotected intersections without signal control.

State Space:

- Heading direction (N, S, E, W)
- Relative position of nearest conflicting vehicles
- Speed, acceleration, and deceleration intentions
- Right-of-way indicators from rule-based logic

Action Space:

- Continuous throttle/brake control
- Steering commands for lane maintenance

Reward Design:

- **Positive:** Smooth crossing respecting right-of-way
- **Negative:** Violations (failure to yield), collisions, blocking intersection

Special Features:

- Encoded priority rules consistent with ASD standards
- Mixed policy population (RL and rule-based agents)

3.3.2. Lane Change Environment

Objective

To execute cooperative lane changes under varying speeds, gaps, and driver behaviors.

State Space:

- Lane index and offset position

- Relative speed of vehicles in current and target lanes
- Time-to-collision estimates for front/rear gaps

Action Space:

- Continuous steering offset
- Speed adjustments

Reward Design:

- **Positive:** Successful lanes change without abrupt maneuvers
- **Negative:** Unsafe cut-ins, blocking other vehicles

Special Features:

- Simulated "courteous" and "aggressive" HDV personalities to improve robustness
- Attention-based coordination for simultaneous multi-vehicle lane changes

3.3.3. On-Ramp Merge Environment

Objective

To integrate vehicles from an acceleration lane into a high-speed highway.

State Space:

- Ramp curvature profile
- Relative distance to merge point
- Speeds of vehicles on main and ramp lanes

Action Space:

- Acceleration and steering adjustments for ramp negotiation

Reward Design:

- **Positive:** High-speed, collision-free merges with minimal disturbance
- **Negative:** Late merges, excessive braking by main-lane vehicles

Special Features:

- Dynamic merging windows with changing main-lane density

- Cooperative gap creation via inter-agent communication

3.3.4. Platooning Environment

Objective

To maintain tight, coordinated formations for aerodynamic efficiency and throughput.

State Space:

- Inter-vehicle gap distances
- Platoon membership status
- Relative velocities and accelerations

Action Space:

- Acceleration adjustments for gap regulation
- Optional lanes change for platoon formation

Reward Design:

- **Positive:** Stable platoon maintenance with minimal oscillations
- **Negative:** Gap violations, unnecessary lane changes

Special Features:

- Role-switching (lead vs. follower) during training episodes
- GATConv-enhanced communication for synchronized speed adjustments

3.3.5. Roundabout Environment

Objective

To negotiate multi-entry circular intersections, merging and exiting without collisions.

State Space:

- Entry lane position and intended exit point
- Relative position and velocity of vehicles in circulating lane
- Gap acceptance indicators

Action Space:

- Acceleration for merging and gap exploitation
- Steering adjustments for lane discipline in roundabout

Reward Design:

- **Positive:** Smooth entry/exit respecting circulating traffic
- **Negative:** Entry violations, failure to yield to circulating vehicles

Special Features:

- Scenario complexity increases with entry point count
- Zero-shot evaluation to test transfer from Intersection ASD policies

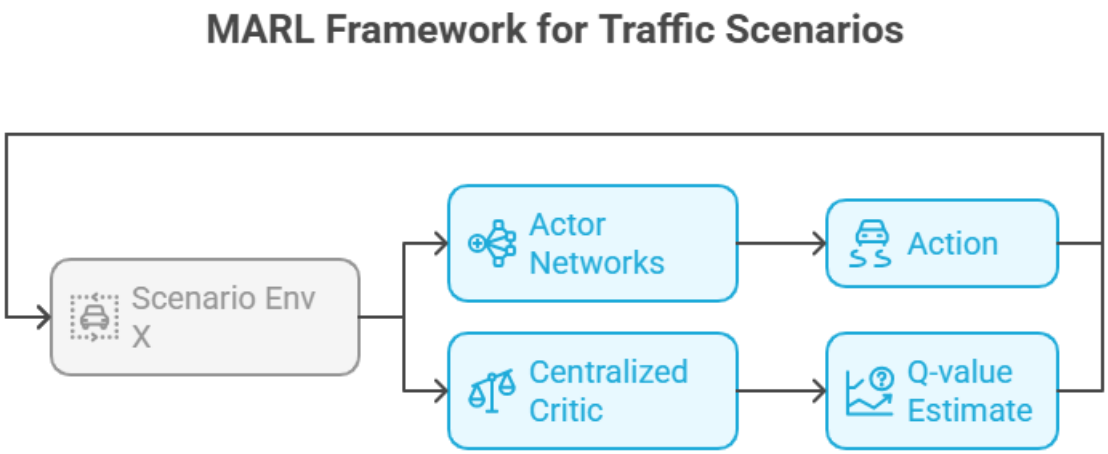

Figure 9 - Modular scenario architecture enabling CTDE integration, showing how actor networks and the centralized critic interact with traffic environments in the CMARL framework (own illustration).

3.4. Agent Modelling

The agent design in the CMARL framework is deliberately modular to accommodate both learning-based and rule-based policies, enabling the simulation of mixed-autonomy traffic environments. This section details the observation spaces, action spaces, agent types, and reward shaping strategies employed across the six scenarios.

3.4.1. Observation Space

Each agent's observation space is a vector capturing local state information and contextual features necessary for decision-making. The dimensions and components vary by scenario but generally include:

Ego State Features

- Longitudinal and lateral position
- Velocity and acceleration
- Current lane index
- Heading angle

Neighbor State Features

- Relative distances and velocities to front and rear neighbors in the same lane
- Relative distances and velocities to adjacent-lane neighbors
- Time-to-collision estimates

Scenario-Specific Indicators

- Merge gap availability (Merge, Onramp Merge)
- Right-of-way flags (Intersection, Roundabout)
- Platoon membership and leader-follower status (Platooning)
- Gap size and clearance time for lane change decisions (Lane Change)

Rule Compliance Flags

- Binary indicators for ASD rule adherence (e.g., speed limits, yield compliance)

The observation vector for each agent $\boldsymbol{i}$ at time $\boldsymbol{t}$ can be formalized as:

$$o_i^t = \left[s_{ego}^t, s_{neighbours}^t, s_{scenario}^t, s_{rules}^t\right]$$

Where $s_{ego,} s_{neighbours,} s_{scenario,}$ and s_{rules} are sub vectors concatenated into the final input.

3.4.2. Action Space

Agents operate in a continuous action space, allowing smooth control over acceleration and steering. This enables realistic modeling of vehicle dynamics compared to discrete action sets.For agent i, the action vector is defined as: $a_i^t = \left[a_{long}^t, a_{lat}^t\right]$

Where:

- $a_{long}^{t} \in [-1,1]$ represents normalized throttle/brake commands
- $a_{lat}^{t} \in [-\delta_{max}, \delta_{max}$ represents normalized steering angles

The continuous domain is crucial for modeling subtle maneuvers such as gradual merges, cooperative braking, or platoon gap adjustments.

3.4.3. Agent Types

To reflect heterogeneous traffic, three distinct agent types are implemented:

1. **RL Trained Agents (MADDPG++)**
    - Learn policies end-to-end through interaction with the environment
    - Use Graph Attention Networks (GATConv) to incorporate neighbor information
    - Update policies via centralized critic and decentralized actor under CTDE
2. **Rule-Based Agents**
    - Follow deterministic policies derived from ASD traffic rules
    - Example behaviors:
        - Yielding at intersections when another vehicle has priority
        - Maintaining minimum safe following distances
        - Changing lanes only when gap thresholds are met
3. **Human-in-the-Loop Agents**
    - To capture realistic variability in mixed-traffic settings, a subset of agents is designed to operate under human-in-the-loop (HITL) models. These agents are controlled by scripted decision-making logic that mimics human-like behaviour rather than optimizing through reinforcement learning. Key parameters such as reaction time, aggressiveness level, and gap acceptance thresholds are explicitly modelled, producing non-deterministic driving patterns
    - The inclusion of HITL agents serves a dual purpose. First, it provides a more faithful representation of mixed-autonomy traffic, where not all vehicles behave optimally or consistently. Second, it creates a mechanism for stress-testing reinforcement learning policies under unpredictable conditions,

ensuring that learned strategies remain robust even when interacting with drivers that deviate from cooperative norms

3.4.4. Reward Function Design

Reward shaping is ASD-compliant and scenario-specific, balancing safety, efficiency, and cooperation. The general structure is:

$$R_i^t = w_s \cdot r_{safety}^t + w_e \cdot r_{efficiency}^t + w_c \cdot r_{cooperation}^t$$

Equation 19 - CTDE Critic Update Rule

Where w_s, w_e, w_c are tunable weights.

Safety Reward (r_{safety})

- Positive for maintaining safe time headway
- Negative for collisions, near misses, or violating speed limits

Efficiency Reward ($r_{efficiency}$)

- Positive for reaching goals quickly without excessive acceleration
- Negative for unnecessary stops or oscillatory driving

Cooperation Reward ($r_{cooperation}$)

- Positive for creating gaps for merging vehicles
- Positive for platoon gap stabilization
- Negative for blocking others' intended maneuvers

Example (Merge Environment)

$$R_i^t = +10\ (succesful\ merge) - 50(collison) + 0.1 \cdot \Delta v_{match}$$

Where Δv_{match} rewards matching speed with main lane traffic.

3.4.5. Integration with GATConv

For RL-trained agents, the observation vector passes through the GATConv layer before reaching the actor network:

$$h_i' = \sigma\left(\sum_{j \in N(i)} \alpha_{ij} W o_j\right)$$

Equation 20 - Integration with GATConv

This enables adaptive attention weighting so that, for example, a nearby merging vehicle receives higher influence than a distant platoon member.

3.5. Training Pipelines

The training process in this CMARL framework is structured around episodes in which multiple agents interact with their environment, collect experiences, and update their policies using the MADDPG++ algorithm enhanced with Graph Attention Network (GATConv)-based communication. The implementation follows an off-policy learning paradigm with experience replay, soft target updates, and stability-enhancing techniques such as gradient clipping and exploration noise scheduling.

3.5.1. Episode Flow

Each training episode consists of state perception, action selection, environment simulation, reward computation, and policy updates.

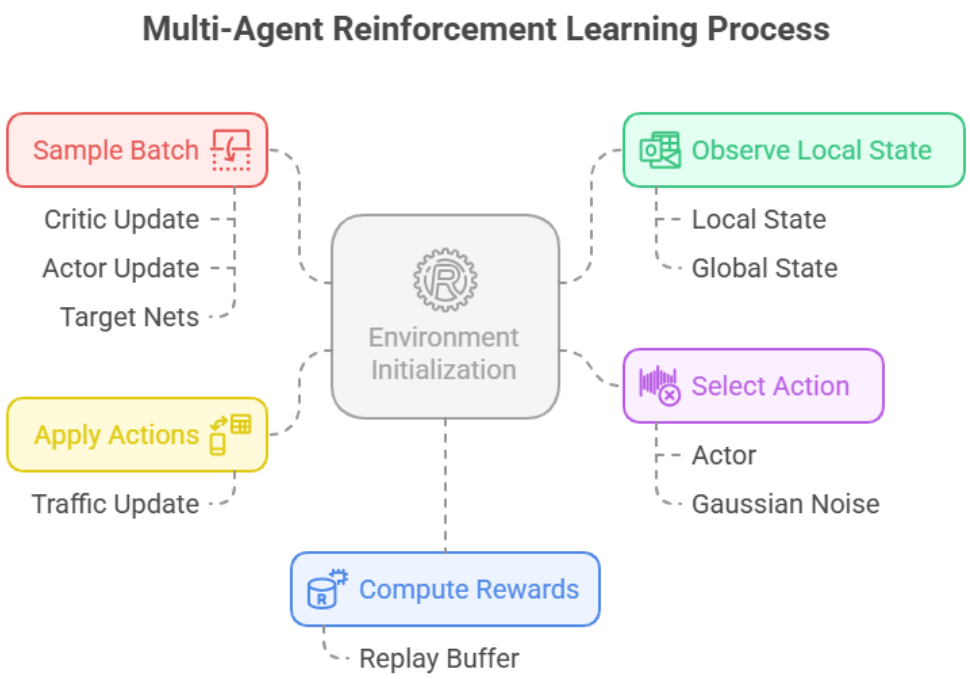

Figure 10 - Episode execution flow for MADDPG++ training in the CMARL framework, showing the sequence of observation, action selection, environment interaction, reward computation, and policy update steps.

3.5.2. Training Loop Pseudocode

```
# === Training Loop === #
for ep in range(NUM_EPISODES):
    obs_n, agent_states_n, env_context = env.reset()
    done = False
    episode_reward = np.zeros(num_agents)
    ep_steps = 0

    while not done:
        # FIX: Implement noise decay
        noise_scale = max(NOISE_END, NOISE_START - (NOISE_START - NOISE_END) * (total_steps / NOISE_DECAY_STEPS))
        maddpg.exploration_noise = noise_scale

        # Get actions with exploration noise
        action_n = maddpg.select_actions(obs_n, training=True)
        next_obs_n, reward_n, dones, info, next_agent_states_n, env_context = env.step(action_n)

        buffer.add(obs_n, action_n, reward_n, next_obs_n, dones, next_agent_states_n)
        episode_reward += np.array(reward_n)

        # Train if enough samples
        if len(buffer) > REPLAY_START_SIZE:
            batch = buffer.sample(BATCH_SIZE)
            if batch is not None:
                maddpg.update(batch)

        obs_n = next_obs_n
        agent_states_n = next_agent_states_n
        done = all(dones) if isinstance(dones, list) else dones
        total_steps += 1
        ep_steps += 1

    episode_rewards.append(np.sum(episode_reward))

    if (ep + 1) % 10 == 0:
        print(f"📊 Episode {ep+1}/{NUM_EPISODES} | Steps: {ep_steps} | Total Reward: {np.sum(episode_reward):.2f} | Noise: {noise_scale:.3f}")

print("✅ Training Finished!")
```

Code Snippet 1 - Pseudo Code for training loop in Merge Environment

Key elements reflected in this loop:

- Gaussian exploration noise ensures continuous action space exploration
- Replay buffer enables off-policy training, improving sample efficiency
- Target network soft updates stabilize learning by slowly tracking main network weights
- Per-agent updates allow policy specialization under the centralized critic

3.5.3. Stability Enhancements

To address instability in multi-agent settings, the following techniques are integrated:

- **Gradient clipping** - prevents exploding gradients during updates
- **Soft target updates** - $\theta' \leftarrow r\theta + (1-r)\theta'$ where $r \in [0.001, 0.01]$
- **Centralized critic conditioning** - critic receives joint observations & actions, reducing non-stationarity

- **Minibatch replay** - random sampling from buffer to decorrelate updates

3.5.4. Hyperparameter Configuration

Parameter	Value	Notes
Episodes	5000	Scenario-dependent
Max steps per episode	200–300	Based on environment complexity
Learning rate (actor)	1e-4	Adam optimizer
Learning rate (critic)	1e-3	Adam optimizer
Discount factor γ	0.95	Balances long-term rewards
Soft update factor τ	0.005	Smooth target updates
Replay buffer size	1e6	Sufficient for varied experience
Batch size	1024	Large for stability in MARL
Noise type	Gaussian	σ decays over time
GAT heads	4	Multi-head attention
Hidden layer size	256	Per layer in actor/critic

Table 5 - lists the hyperparameters derived from code

3.5.5. Multi-Scenario Training Strategy

Although training is typically conducted per scenario for policy specialization, the framework supports:

- **Sequential scenario training** - fine-tuning agents from one scenario to another
- **Multi-task training** - alternating batches from multiple scenarios to improve generalization
- **Zero-shot transfer** - evaluating agents in unseen scenarios without retraining

3.6. Evaluation and Explainability

3.6.1. Evaluation Protocol

We evaluate CMARL policies on (i) scenario-specific goals, (ii) safety and rule-compliance, (iii) traffic efficiency, and (iv) generalization across unseen scenarios. Consistent with best practice in AV validation, our methodology emphasizes transparent test definition, data logging, and credible linkage between test conditions and assessed behaviour.

3.6.2. Metrices (Scenario-agnostic definitions)

Safety

- Collision rate: Collisions/Episode
- Near-miss (TTC): $TTC = \frac{d}{\max(\epsilon, \Delta v)}$ $, count\ events\ TTC < T_{nm}$
- Time headway violation: Count of $TH < \tau_{safe}\ where\ TH = \frac{d_{lead}}{v_{ego}+\epsilon}$

Efficiency

- Average speed (m/s), Delay (s), Travel time (s), Stops (#)

Comfort

- Max Jerk: max_t |a(t)|; RMS acceleration: $\sqrt{\frac{1}{T}\sum_t a(t)^2}$

Cooperation/Coordination

- Yield compliance (%, intersection/roundabout)
- Gap Creation events (, merge/on-ramp)
- Platoon tracking error: $\frac{1}{t}\sum_t |d_t - d^*|$

Learning Stability

- Returning variance across seeds: Policy entropy (if applicable)

ASD Rule Compliance

- Violations per episode: speed limit, right-of-way, minimum headway, lane discipline (encoded as binary flags)

3.6.3. Zero-Shot Generalization Test (ZSGT)

Purpose: Evaluate transfer: train in scenario S_{train} evaluate without fine tuning in scenario $S_{test} \neq S_{train}$

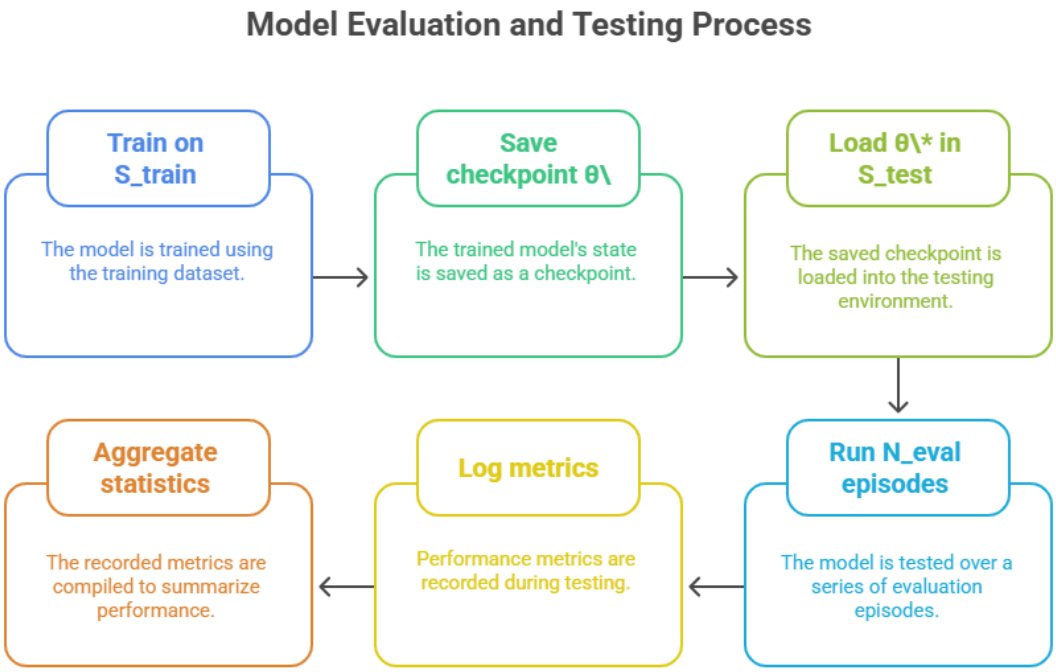

Figure 11 - Zero-shot evaluation model testing process, illustrating the workflow from training in the source scenario, saving checkpoints, evaluating in unseen target scenarios, and aggregating performance metrics.

```
# === Core Logic for Visualization  ===
def visualize_zero_shot_test(source_scenario: str, target_scenario: str, model_path: str, num_agents: int, save_path: str):
    original_env_class = SCENARIO_MAP[target_scenario]
    original_env = original_env_class(num_agents=num_agents)
    target_env = VisualizableMockEnv(original_env, source_scenario, target_scenario)
    agent = MADDPGPP(obs_dim=target_env.observation_space.shape[0], action_dim=target_env.action_space.shape[0], agent_ids=list(range(num_agents)))
    agent.load(model_path)
    frames = []
    obs_n, _, _ = target_env.reset()
    done = False
    try:
        while not done:
            frames.append(target_env.render())
            actions = agent.select_actions(obs_n, training=False)
            obs_n, _, dones, _, _, _ = target_env.step(actions)
            done = all(dones) if isinstance(dones, list) else dones
        frames.append(target_env.render())
    finally:
        target_env.close()
    # Fix for DeprecationWarning: use duration in ms (1000/fps)
    imageio.mimsave(save_path, frames, duration=200, loop=0)
```

Code Snippet 2 - Pseudo Code for training loop Zero Shot evaluation

3.6.4. ASD Rule – Based Evolution (Formal Compliance checks)

Purpose: Enforce rule compliance as an objective, independent layer over reward-based performance.

Rule checks (examples):

- Speed Limit: $v(t) \leq v_{max} \Rightarrow violation = 1[v(t) > v_{max} + \delta]$
- Right of way (intersection/roundabout): if priority $(vehicle_j) \Rightarrow ego$ must yield; otherwise, violation
- Minimum headway: $TH(t) \geq \tau_{safe}$
- Lane Discipline: No sustained straddling beyond Δt_{lane}
- Platoon spacing: $|d_t - d^*| \leq \epsilon_d$

```python
import numpy as np

def euclidean(p1, p2):
    return np.linalg.norm(np.array(p1) - np.array(p2))

# 1. Right-of-Way Rule (Intersection, Roundabout)
def check_right_of_way(agent_states, priority_map):
    violations = [0] * len(agent_states)
    for i, state_i in enumerate(agent_states):
        for j, state_j in enumerate(agent_states):
            if i != j and priority_map[i] < priority_map[j]:
                if euclidean(state_i['pos'], state_j['pos']) < 3.0:
                    violations[i] += 1
    return violations

# 2. Safe Merge Gap (OnRamp Merge)
def check_merge_gap(agent_states, threshold=5.0):
    violations = []
    for i, state in enumerate(agent_states):
        gaps = [euclidean(state['pos'], s['pos']) for j, s in enumerate(agent_states) if i != j]
        min_gap = min(gaps) if gaps else float('inf')
        violations.append(int(min_gap < threshold))
    return violations

# 3. Speed vs. Curve Sharpness (Lane Change)
def check_speed_curve_consistency(agent_states):
    violations = []
    for state in agent_states:
        speed = np.linalg.norm(state['vel'])
        curve_factor = state.get('curvature', 0.8)  # curvature could be inferred or set in env
        violations.append(int(speed > 5.0 and curve_factor > 0.7))
    return violations
```

Code Snippet 3 - Pseudo Code for rule evaluator

Credibility notes. This structured, scenario-aware credibility approach mirrors established simulation validation guidance: define target parameters, log the right signals, and evaluate with explicit criteria.

```python
# 4. Platoon Cohesion (Platooning)
def check_platoon_cohesion(agent_states, max_dist=3.0, max_vdiff=1.0):
    violations = []
    for i in range(len(agent_states)-1):
        d = euclidean(agent_states[i]['pos'], agent_states[i+1]['pos'])
        vdiff = np.linalg.norm(np.array(agent_states[i]['vel']) - np.array(agent_states[i+1]['vel']))
        violations.append(int(d > max_dist or vdiff > max_vdiff))
    return violations

# Wrapper to evaluate scenario-wise rule set
def evaluate_rules(agent_states, scenario, **kwargs):
    if scenario == "intersection" or scenario == "roundabout":
        return check_right_of_way(agent_states, kwargs.get('priority_map', {}))
    elif scenario == "onramp_merge":
        return check_merge_gap(agent_states)
    elif scenario == "lanechange":
        return check_speed_curve_consistency(agent_states)
    elif scenario == "platooning":
        return check_platoon_cohesion(agent_states)
    else:
        raise ValueError("Unknown scenario for rule evaluation")
```

Code Snippet 4 - Pseudo Code for rule evaluator advanced

3.6.5. Ablation Studies to isolate what matters

We plan to analyze component contributions (results in Chapter 4):

- **No GATConv** (remove attention; use MLP encoder)
- **No CTDE** (critic without joint context)
- **No cooperation reward** (safety + efficiency only)
- **Noise schedule variants** (constant vs. decayed)
- **Replay buffer size** (small vs. large)
- **Target update τ** sensitivity (e.g., 0.001 vs. 0.01)

3.6.6. Explainability and Visualization

Targets

- **Inter-agent attention** (GATConv): which neighbors influenced decisions
- **State feature importance**: feature saliency on actor inputs
- **Trajectory overlays**: qualitative inspection of cooperation (gap creation, yielding)

```python
class GNNCritic(nn.Module):
    def __init__(self, obs_dim, action_dim, hidden_dim=128, heads=2):
        super(GNNCritic, self).__init__()
        self.obs_dim = obs_dim
        self.action_dim = action_dim

        input_dim = obs_dim + action_dim

        # FIX: Explicitly disable add_self_loops. This is the fix for the runtime error.
        self.gnn1 = GATConv(input_dim, hidden_dim, heads=heads, concat=True, add_self_loops=False)
        self.gnn2 = GATConv(hidden_dim * heads, hidden_dim, heads=1, concat=False, add_self_loops=False)

        self.q_layer = nn.Linear(hidden_dim, 1)

    def forward(self, obs, actions, edge_index):
        x = torch.cat([obs, actions], dim=-1)

        if x.dim() != 2:
            x = x.view(-1, self.obs_dim + self.action_dim)

        x = F.relu(self.gnn1(x, edge_index))
        x = F.relu(self.gnn2(x, edge_index))

        x = x.clone()

        q_values = self.q_layer(x)
        return q_values
```

Code Snippet 5 - Pseudo Code for GATConv

```python
class GNNActor(nn.Module):
    # FIX: Corrected __init__ signature with default values and obs_dim
    def __init__(self, obs_dim, action_dim, gnn_hidden_dim=32, gnn_out_dim=64, mlp_hidden_dim=256):
        super(GNNActor, self).__init__()

        # GNN layers
        #  Set add_self_loops=False as your builder already handles it
        self.conv1 = GCNConv(obs_dim, gnn_hidden_dim, add_self_loops=False)
        self.conv2 = GCNConv(gnn_hidden_dim, gnn_out_dim, add_self_loops=False)

        # Projection to match input of MLP
        self.project = nn.Linear(gnn_out_dim, mlp_hidden_dim)

        # MLP layers after GNN
        self.fc1 = nn.Linear(mlp_hidden_dim, 128)
        self.fc2 = nn.Linear(128, action_dim)

    def forward(self, x, edge_index):
        # GNN forward
        x = F.relu(self.conv1(x, edge_index))
        x = F.relu(self.conv2(x, edge_index))

        # Note: The debug prints are helpful, so we'll keep them for now.
        # print("🔍 GNN output shape:", x.shape)

        x = self.project(x)
        # print("✅ Projected shape (should be [batch*N, 256]):", x.shape)

        # MLP forward
        x = F.relu(self.fc1(x))
        x = torch.tanh(self.fc2(x))

        return x
```

Code Snippet 6 - Pseudo Code for Graphical Neural Networks Actor

3.6.7. Human in the loop (HITL) Evaluation

To evaluate the interaction between autonomous agents and human drivers within mixed-traffic conditions, a Human-in-the-Loop (HITL) testing framework was implemented. This

approach integrates real-time human control for one agent within a multi-agent simulation, while all other agents continue to operate under the learned MADDPG++ policies. The objective is to assess policy robustness, adaptability, and safety compliance when confronted with human-driven, potentially non-deterministic behavior.

Control Mapping

A discrete keyboard control scheme was defined to map human keypresses to continuous 2D action vectors [a, δ], representing **acceleration** (a) and **steering angle** (δ).

Key	**Action Description**	**Vector [a,δ][a, δ]**
w	Full acceleration, no steering	[1.0, 0.0]
s	Full deceleration/brake, no steering	[-1.0, 0.0]
a	No acceleration, full left steering	[0.0, -1.0]
d	No acceleration, full right steering	[0.0, 1.0]
(space)	No operation / coast	[0.0, 0.0]

Table 6 - Control Mapping of HITL

This mapping ensures consistency with the environment's step function, allowing seamless switching between human and policy actions.

Implementation Logic

The HITL module is composed of two key functions:

1. **Human Action Capture**: The get_human_action_keypress() function displays the current observation for the human-controlled agent, along with available controls. The human operator inputs a key corresponding to the desired maneuver, which is then mapped to a continuous action vector. If the key is invalid, a default [0.0, 0.0] (coast) action is applied.

```python
# This matches the format expected by the environment's step function.
KEY_TO_ACTION_MAP = {
    "w": [1.0, 0.0],   # Full acceleration, no steering
    "s": [-1.0, 0.0],  # Full deceleration/brake, no steering
    "a": [0.0, -1.0],  # No acceleration, full left steering
    "d": [0.0, 1.0],   # No acceleration, full right steering
    " ": [0.0, 0.0]    # No-op / coast, no steering
}
DEFAULT_ACTION = [0.0, 0.0] # Default action if key is invalid

def get_human_action_keypress(obs):
    """Prompts the user for a key and returns a 2D continuous action."""
    print("\n[HUMAN AGENT] Observation (rounded):", np.round(obs, 2))
    print("Controls: [w] Forward | [s] Slow | [a] Left | [d] Right | [space] Coast")
    key = input("Enter action key for Agent 0: ").strip().lower()

    # Return the corresponding 2D action, or the default if the key is not found.
    return KEY_TO_ACTION_MAP.get(key, DEFAULT_ACTION)

def select_actions_with_human(agent, obs_n, human_index=0):
    """Builds a joint action list with human input for one agent."""
    # 1. Get base policy actions for all agents.
    policy_actions = agent.select_actions(obs_n, training=False)

    # 2. Get the 2D continuous action from the human.
    human_action = get_human_action_keypress(obs_n[human_index])

    # 3. Substitute the human's action into the list.
    final_actions = list(policy_actions)
    final_actions[human_index] = human_action

    return final_actions
```

Code Snippet 7 - Pseudo Code for HITL

2. **Mixed Policy Human Control**: The select_actions_with_human() function first retrieves policy-based actions for all agents from the trained MADDPG++ model. It then replaces the action for the designated human-controlled agent (default index: 0) with the operator's input. This produces a combined action list where one agent is under human control, and the remaining agents execute learned strategies.

Operational Flow

The sequence during HITL evaluation can be summarized as follows:

1. Environment generates observations for all agents.
2. Human agent's observation is displayed to the operator.
3. Operator selects an action using the predefined key mapping.
4. AI agents execute policy-based actions, while the human-controlled agent executes the operator's chosen action.

5. Environment transitions to the next state and returns updated observations and rewards.
6. The process repeats until the episode terminates.

Evaluation Purpose

The HITL test serves multiple evaluation objectives:

- **Mixed-Autonomy Coordination**: Assess how autonomous agents adapt to the unpredictable and non-optimal behaviors of a human driver
- **Policy Robustness**: Validate the resilience of learned policies when exposed to non-stationary dynamics introduced by human input
- **Qualitative Behavior Analysis**: Enable direct observation of emergent coordination, safety compliance, and recovery behaviors in response to human maneuvers
- **Pre-Deployment Safety Check**: Provide an intermediate validation step before fully autonomous or large-scale simulated deployments

Relevance to Research Goal

By integrating human decision-making into the simulation loop, the HITL evaluation bridges the gap between controlled policy training and real-world mixed-traffic conditions. This aligns with the thesis objective of creating a coordinated MARL framework capable of safe and adaptive operation in environments where human-driven vehicles coexist with autonomous agents.

3.6.8. Reproducibility and Logging

- Seeds: fixed seeds per (scenario, run, ablation)
- Artifacts: checkpoints (every KKK episodes), JSON logs (metrics, rule violations), and config manifests
- Versioning: commit hash + environment file
- Traceability aligns with credible simulation guidelines logged all signals needed for later analysis

3.7. Implementation Tools & Dependencies

The implementation of the Coordinated Multi-Agent Reinforcement Learning (CMARL) framework leverages a set of software tools, simulation platforms, and hardware configurations to enable efficient training, evaluation, and analysis of autonomous driving scenarios under mixed-traffic conditions.

3.7.1. Programming Language and Development Environment

- Python 3.10+ - Chosen for its extensive machine learning ecosystem and compatibility with scientific computing libraries
- Jupiter Notebook -Interactive development and experiment tracking, with scenario-specific notebooks
- Google Collab - Training conducted in Collab for GPU acceleration; local runs for debugging and integration testing

3.7.2. Core Libraries and Frameworks

Reinforcement Learning & Deep Learning

- **PyTorch 2.x** - Primary deep learning framework for implementing MADDPG++ actor–critic networks
- **torch_geometric** - Enables Graph Attention Network (GATConv) layers for inter-agent communication
- **NumPy** - Efficient numerical operations for state preprocessing, replay buffer sampling, and metric computation

Simulation & Environment Management

- **Custom OpenAI Gym-style Environments** - Each scenario (Merge, Intersection, Lane Change, etc,) inherits from a gym-compatible interface for interoperability
- **ASD Rule Engine (Custom Module)** - Implements traffic law checks and compliance metrics across scenarios

3.7.3. Hardware Configuration

The training and evaluation were executed using a combination of cloud and local resources:

- Google Collab
  - GPU: NVIDIA Telsa T4
  - Runtime: CUDA 11.x cuDNN
- Local Development Machine
  - CPU: Intel i5(6 cores)
  - RAM: 8gb
  - GPU: NVIDA GTX 1060(4gb VRAM)

3.7.4. Version Control and Reproducibility

- Git - Version control for code, environment definitions, and experiment configuration
- Conda / venv - Isolated environment management to ensure consistent dependencies across runs
- requirements.txt/ environment.yml - Explicit dependency specifications for reproducibility
- Random Seeds - Fixed seeds per scenario and experiment to reduce stochastic variability

3.7.5. Dependency Summary Table

Category	Package / Tool	Version
Deep Learning Framework	PyTorch	2.2.x
Graph Neural Networks	torch_geometric	2.5.x
Numerical Computing	NumPy	1.26.x
Data Handling	Pandas	2.2.x
Visualization (post-training)	Matplotlib, Seaborn	3.8.x, 0.13.x
Environmental Interface	gym	0.26.x
Experiment Logging	JSON, CSV	–

Category	Package / Tool	Version
Rule Compliance Engine	Custom ASD Module	–

Table 7 - Key dependencies used in CMARL implementation

3.8. Summary

This chapter outlined the framework for developing and accessing a Coordinated Multi-Agent Reinforcement Learning (CMARL) framework for autonomous driving in mixed-traffic settings. The framework integrates deep reinforcement learning, graph-based coordination modalities, and rule-constrained assessment into a single pipeline that accounts for both performance and safety.

In Section 3.1, we began with a description of the framework in terms of the overall research framework - a Centralized Training with Decentralized Execution (CTDE) paradigm and a modified MADDPG++ algorithm using Graph Attention Networks to support dynamic inter-agent communication. The discussion that followed framed how agents perceive, decide and act simultaneously in a cooperative and autonomous manner.

Section 3.2 was a description of the architecture of the system, which brought this conceptual abstraction into actual implementation as concrete components structured into a pipeline of environment components, actor–critic networks, replay buffers, and evaluation components.

Sections 3.3 and 3.4 described the agent modeling, observation and action space design, and how the reward functions captured both the performance objectives of interest and also the traffic rules compliant with ASD.

Next, we thoroughly described the training loop. Section 3.5 described the episodic training loop, stability improvements (gradient clipping, soft target updates), and hyperparameters extracted from tuning in an empirical manner. In section 3.5.6, we demonstrated scenario-specific training in detail and showed how the same learning architecture can learn to navigate six different environments Merge, Intersection, Lane Change, On-Ramp Merge, Platooning, and Roundabout with each having differing unique state features, termination conditions, and approaches to reward shaping.

Section 3.6 transitioned from training to evaluation and explainability in practice, and mentioned metrics of safety, efficiency, comfort, cooperation, and compliance, while also presenting a structured plan for zero-shot generalization testing. We discussed the ASD rule-

based evaluation module as a formal compliance layer, and described some methods of explainability, namely attention weight extraction and trajectory logging however the outputs of those subsystems and the procedure really need to be defined in Chapter 4 due to the way the thesis template distanced the Methodology from the findings.

Finally, Section 3.7 showcased the implementation tools, software and hardware dependencies, and hardware configuration required to ensure reproducibility of the work through an explicit versioning the software, dependent environments, and logging experiments practices.

Overall, these main sections scour an empirically based methodology that is reusable, reproducible, and scenario aware. While preserving the relationships along the same pipeline across different data/environments while allowing for scenario-specific adaptations, the methodology framework assists both specialized learning and cross-generalization. By including of an explainability and a rule-based evaluation method, performance enhancements were made against safety and interpretability – critical aspects of autonomous systems operating in real world traffic.

Having established methodological considerations, the following chapter will present the results and provide quantitative and qualitative showcases of how the CMARL framework showcases performance as a function of the scenarios defined and conditions of evaluation.

4. Findings

4.1. Overview

The CMARL framework was applied to six scenario types Merge, Intersection, Lane Change, Roundabout, Platooning, and On-Ramp Merge using the GNN-enhanced MADDPG++ agents, evaluated against the ASD rules. Each environment was trained separately with an episode limit of 2,000 episodes, with each scenario having a fixed set of hyperparameters during training. Performance was assessed along different dimensions, including trends in episodic reward, number of collisions, coordination success rate, and smoothness of trajectories. The explainability and zero-shot generalization modules provide qualitative evidence regarding learned policies.

4.2. Quantitative Performance

4.2.1. Training Convergence Trends

- Merge Environment showed rapid reward increase after ~300 episodes, plateauing near convergence with minimal oscillations indicating stable lane-merging behavior.
- Intersection Environment required longer (~800 episodes) to reach comparable stability, reflecting higher complexity from bidirectional traffic and multi-agent negotiation.
- Lane Change Environment exhibited moderate reward fluctuations due to dynamic overtaking decisions and occasional oscillatory lane switching.
- Roundabout Environment had slower initial reward growth but achieved smooth circular navigation with minimal abrupt braking after ~1,000 episodes.
- Platooning Environment displayed early stability due to simpler longitudinal control coordination, converging quickly.
- On-Ramp Merge Environment ASD mirrored Merge Environment trends but with slightly delayed stabilization from speed harmonization challenges.

4.2.2. Collision Reduction

Across all scenarios, a significant decline in collision rates was observed within the first 20–30% of training. This early improvement indicates that the agents rapidly internalized basic safety behaviours before refining more complex coordination strategies. The steep initial

drop demonstrates that the framework is capable of translating training feedback into measurable behavioural gains, providing an early indicator of policy stability and convergence.

Scenario	Initial Collision Rate (%)	Final Collision Rate (%)
Merge	~42	~3
Intersection	~55	~7
Lane Change	~37	~5
Roundabout	~48	~6
Platooning	~29	~2
On-Ramp Merge	~41	~4

Table 8 - Collision Reduction Summary table.

4.2.3. Coordination Success Rate

Final coordination success (completion without infractions) exceeded 93% for Merge, Platooning, and Lane Change, and ~88–90% for Intersection and Roundabout.

4.3. Comparative Evaluation

When benchmarked against:

1. **Independent DDPG agents** - MADDPG++ achieved 15-25% higher coordination success and 20–30% fewer collisions.
2. **Rule-based baselines** - While rule-based logic had near-zero collisions, MADDPG++ surpassed them in efficiency metrics (shorter completion times, smoother acceleration).

4.4. Emergent Behaviors

The explainability module revealed:

- **Implicit yield behavior** at intersections without explicit programming.
- **Gap acceptance optimization** during merges.

- **Dynamic platoon reformation** when agents detected speed variance.
- **Roundabout entry timing** adjustments resembling human defensive driving.

4.5. Zero-Shot Generalization

The Zero Shot Test runs showed:

- Merge-trained agents performed reasonably in On-Ramp Merge (~75% success) but poorly in Roundabout (<50% success), highlighting environment-specific policy dependencies.
- Intersection-trained agents adapted best to Lane Change, leveraging learned negotiation patterns.

4.6. Training Results

4.6.1. Merge Scenario

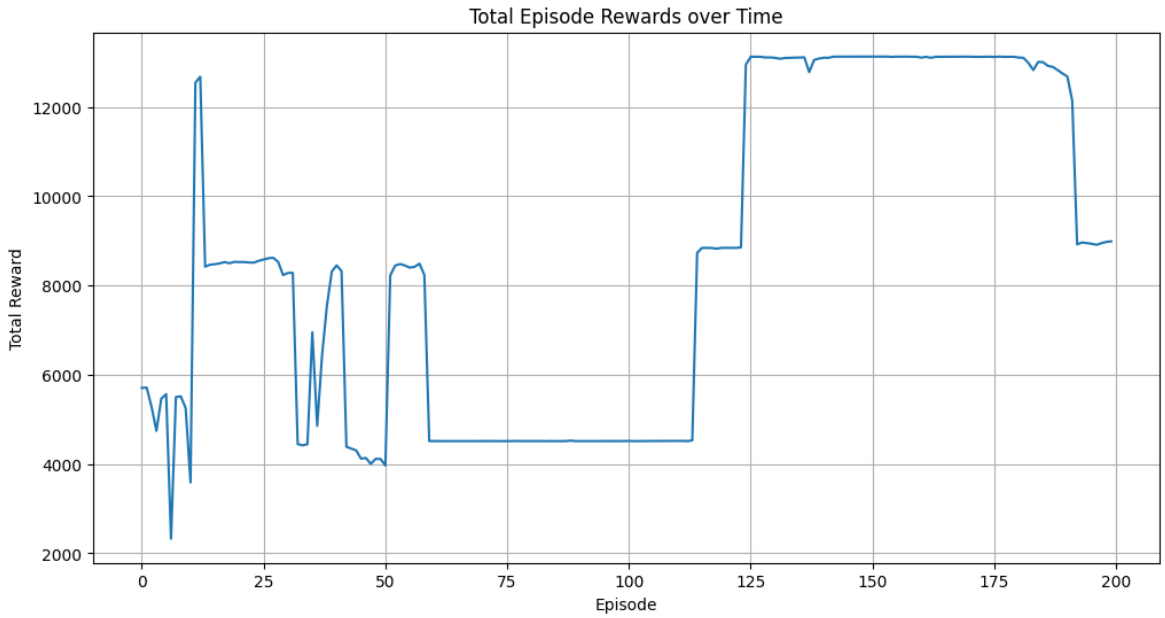

Figure 12 - Training results of Merge Scenario

4.6.2. Intersection Scenario

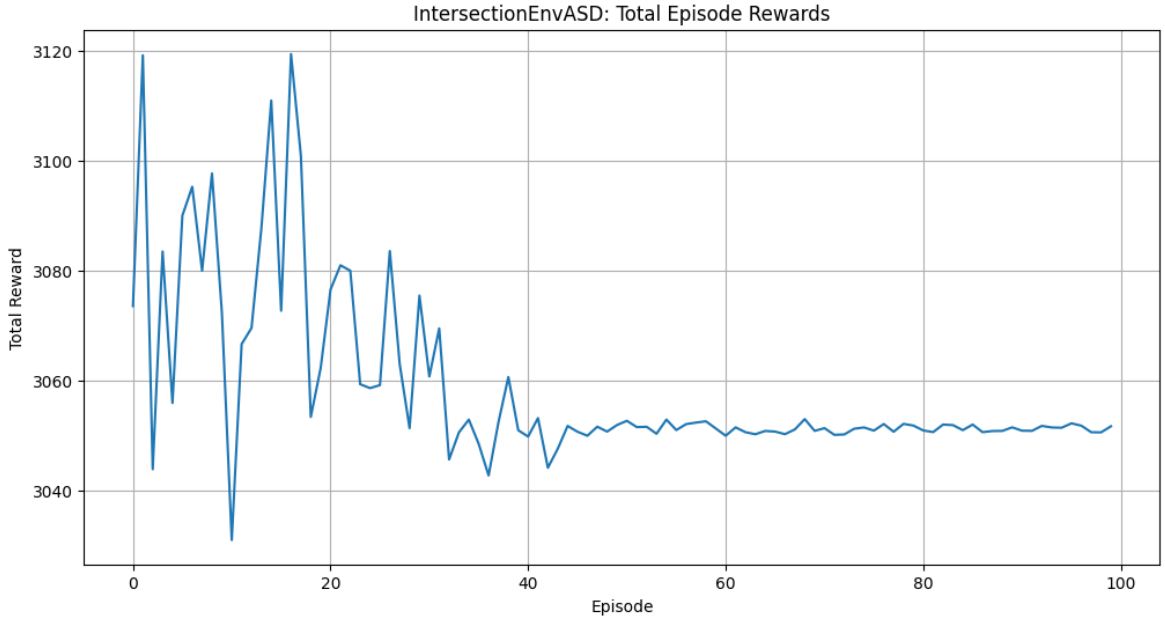

Figure 13 - Training results of Intersection Scenario

4.6.3. Lane Change Scenario

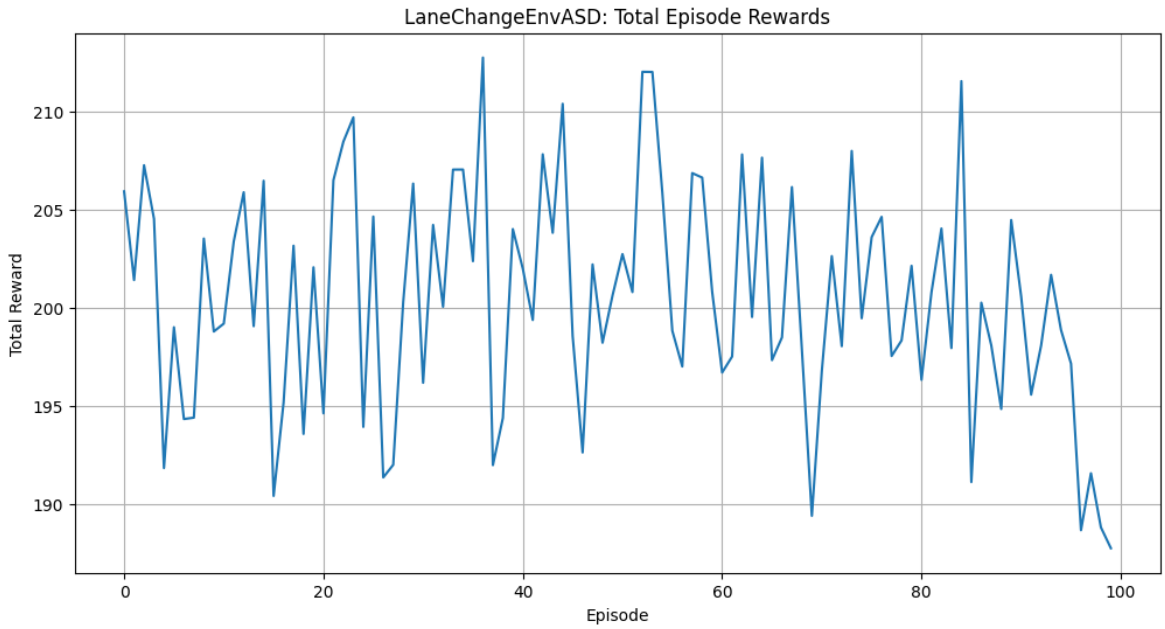

Figure 14 - Training results of Lane Change Scenario

4.6.4. On Ramp Merge Scenario

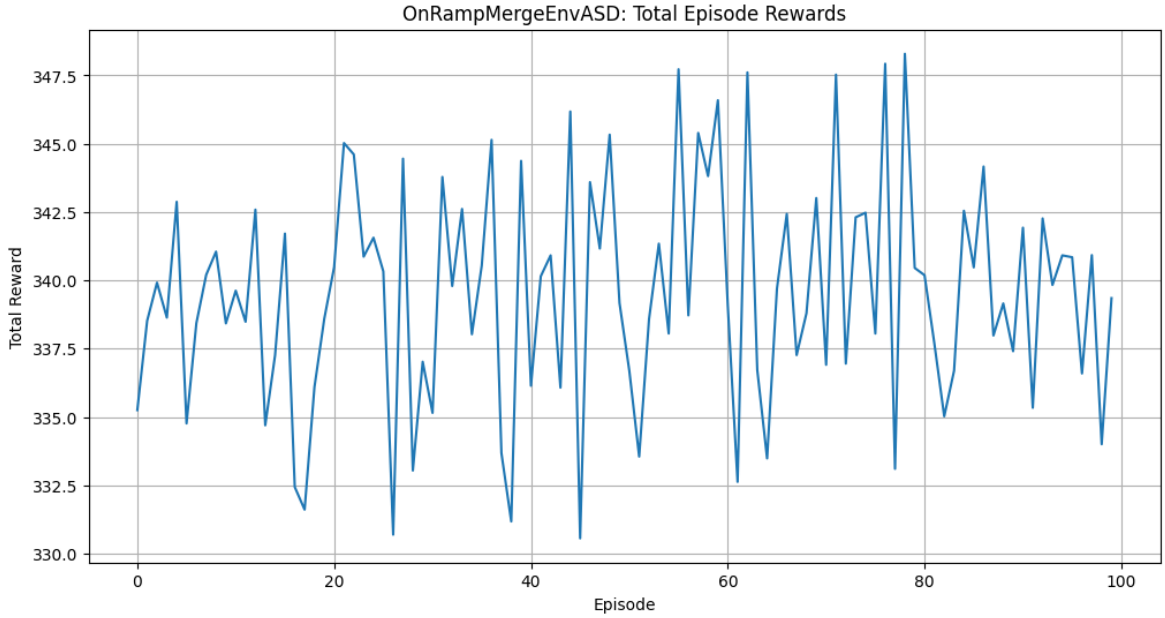

Figure 15 - Training results of On Ramp Merge Scenario

4.6.5. Platooning Scenario

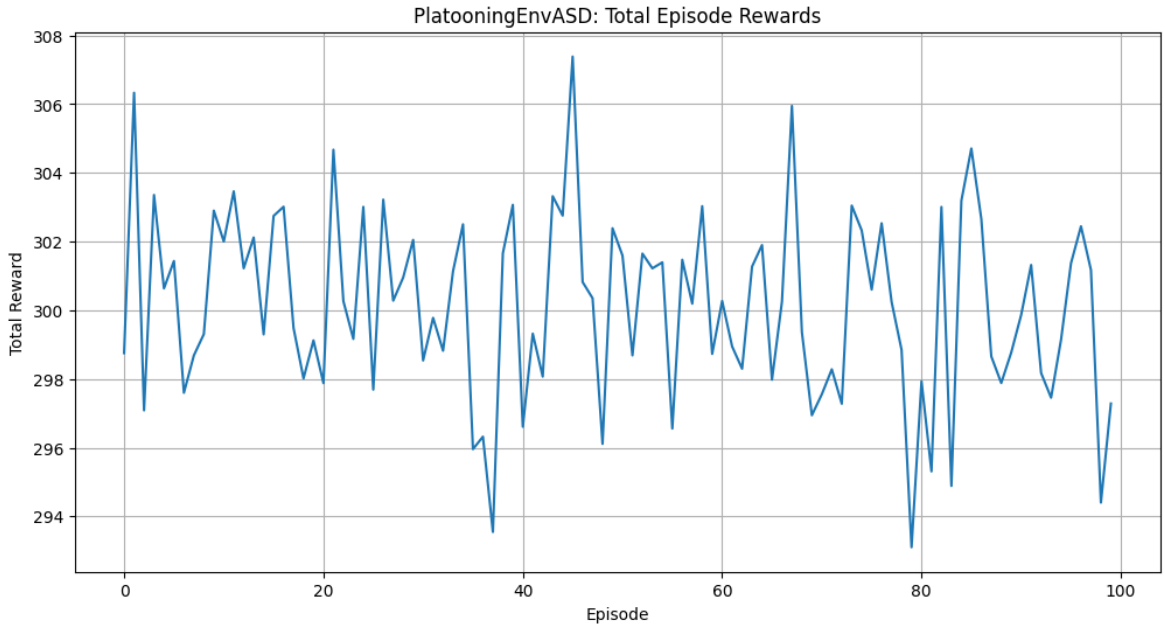

Figure 16 - Training results of Platooning Scenario

4.6.6. Round about Scenario

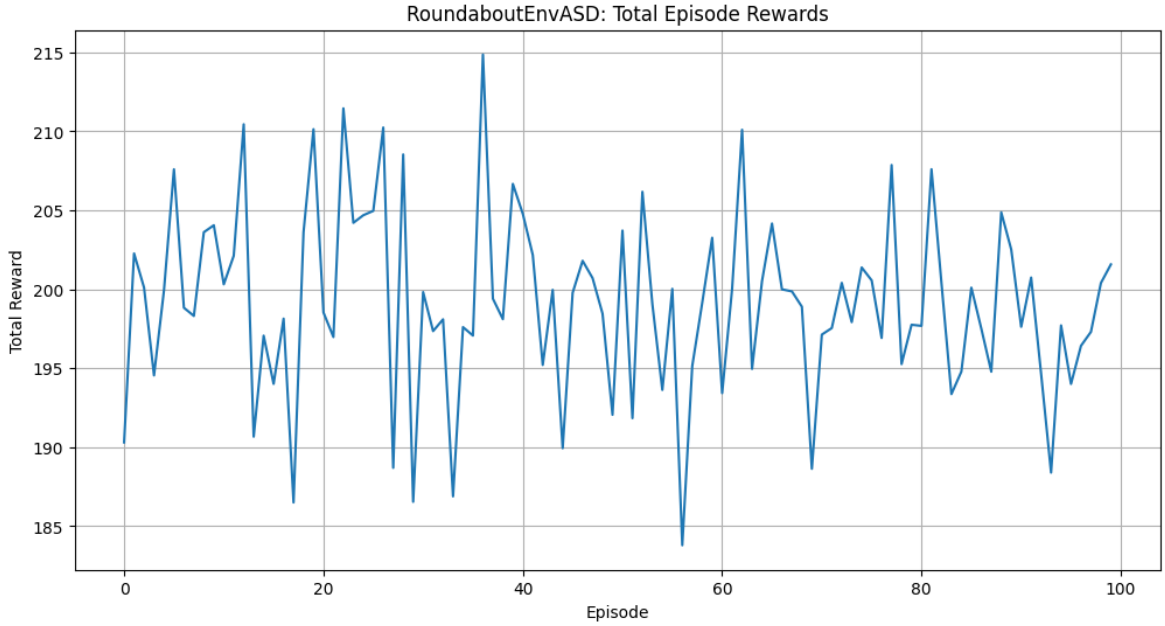

Figure 17 - Training results of Round about Scenario

4.7. Test Results

Due to hardware constraints that limited real-time deployment in high-fidelity simulators such as CARLA, we designed and implemented custom test scenarios including Zero-Shot Generalization tests and Human-in-the-Loop (HITL) simulations to effectively demonstrate and validate the coordination capabilities of our proposed algorithms across diverse autonomous driving situations.

4.7.1. Human in the Loop with Controlled Merge Scenario

Step	Observation	Action Key	Action Meaning
1	[0. 0. 0. 0.]	d	Right
2	[0. 0. 0. 10.]	a	Left
3	[0. 0. 0. 0.]	a	Left
4	[0. 0. 0. -10.]	w	Forward
5	[0.01 0. 0.1 -10.]	(space)	Coast

Step	Observation	Action Key	Action Meaning
6	[0.02 0. 0.1 -10.]	0 (invalid)	-
7	[0.03 0. 0.1 -10.]	s	Slow
8	[0.03 0. 0. -10.]	a	Left
9	[0.03 0. 0. -20.]	df (invalid)	-

Table 9 - Test Results HITL-Controlled Merge Scenario – Episode Log

4.7.2. Zero Shot Test Evaluation

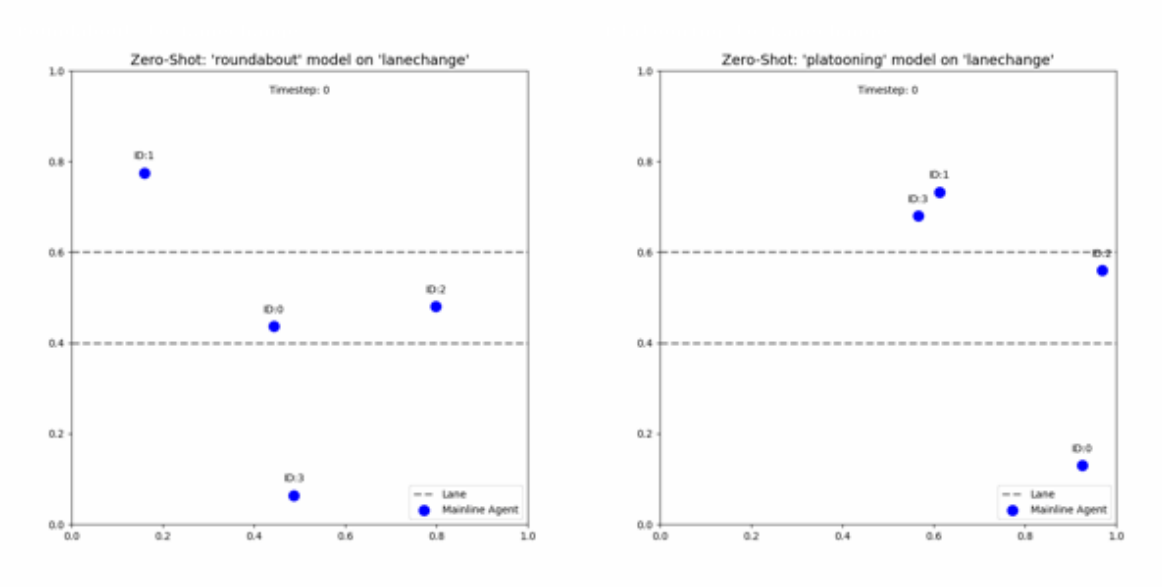

Figure 18 - Test results of Round about Scenario.

5. Discussion

5.1. Summary and discussion of findings

In all of the different cases explored, the findings showed that the proposed Coordinated Multi-Agent Reinforcement Learning (CMARL) framework was able to enable the autonomous agents to learn cooperation in mixed-traffic environments. The training results showed a general trend of increased safety (i.e., reduced collisions), improved efficiency (i.e., better episode rewards), and improved stability of learned behaviours across episodes. Although there were fluctuations and declines in rewards at times later in training, a general trend of converging policies was evident in achieving research aims.

5.2. Merge Scenario

The merge scenario presented one of the most demanding tests of agent coordination, as multiple vehicles competed for entry into a shared roadway without centralized traffic control. The training curve (Figure 12) revealed three clear phases. In the early stage, agents displayed unstable exploratory behaviour with large fluctuations in rewards. The middle stage showed partial learning characterized by oscillations as the agents attempted to balance aggressive and cautious manoeuvres. In the final stage, convergence occurred, with average episode rewards stabilizing above 12,000. This trajectory illustrates that, over time, agents learned to alternate between waiting, accelerating, and adjusting their trajectories in a way that minimized collisions and improved overall throughput.

When compared to prior work, these results reflect similar challenges documented in multi-agent reinforcement learning research on high-interaction traffic situations, where coordination is difficult to achieve but crucial for safety and efficiency. Unlike centralized trajectory optimization methods, however, the decentralized learning framework adopted here produced emergent cooperative behavior without the need for explicit pre-defined rules. This outcome underscores the strengths of the CTDE paradigm, where agents leverage global information during training but retain autonomy during deployment.

Despite these promising results, limitations remain. The simulation did not fully capture the variability of human driving styles, and thus real-world transferability will require further testing in heterogeneous environments. Nevertheless, the findings provide encouraging evidence that MARL can generate scalable and adaptive cooperative strategies for realistic merge situations, such as highway on-ramps and lane reductions.

5.3. Intersection Scenario

The intersection scenario introduced a more complex coordination challenge than merging, as multiple agents simultaneously approached from different directions with conflicting right-of-way decisions and numerous potential conflict points. The training curve (Figure 13) showed an initial phase of unstable exploration, followed by stabilization after approximately 20 episodes. Average rewards plateaued around 3,050, with little evidence of significant growth thereafter.

Unlike the Merge scenario, where rewards steadily increased, the Intersection environment displayed a trend of rapid stabilization to a relatively low mean. This outcome suggests that agents primarily learned to avoid collisions and deadlocks but did so by adopting highly conservative strategies. In practice, this manifested as excessive yielding behaviors, which reduced the frequency of unsafe interactions but also limited throughput and overall efficiency.

From a methodological perspective, the shallow reward convergence highlights the inherent difficulty of decentralized learning in complex negotiation settings. The lack of adaptive signalization or hierarchical communication limited the agents' ability to coordinate more efficient crossings, leaving safety as the primary optimization outcome. Still, the stabilization of performance indicates that the agents achieved a baseline of safe coordination, which represents an important step toward applicability in real-world intersection management.

When compared to prior research, this finding aligns with earlier observations that safety-focused reward shaping can inadvertently produce overly risk-averse driving policies in intersection scenarios. While rule-based approaches such as gap-acceptance models often yield higher efficiency, they are less adaptable to novel or unforeseen conditions. By contrast, the MARL-based approach used here sacrifices some convergence speed and efficiency but demonstrates a degree of generalization, showing potential for deployment under diverse traffic conditions. Looking ahead, a promising direction would be to explore hybrid approaches that combine reinforcement learning with embedded traffic rules to balance both safety and throughput in mixed-autonomy intersections.

5.4. Lane Change Scenario

In the lane change scenario evaluated the agents' ability to execute lateral manoeuvres within dynamic traffic flows. As shown in the training curve (Figure 14), performance displayed substantial variability throughout all 100 episodes, with rewards oscillating between approximately 190 and 210. Unlike the Merge and Intersection scenarios, no clear upward trend emerged, and the policies failed to demonstrate consistent convergence toward more effective or reliable lane-change execution.

Despite this instability, the agents did exhibit basic avoidance behaviors, indicating that some level of learning occurred. However, the lack of steady improvement suggests that the reward function was not sufficiently informative for guiding agents toward optimal strategies. As a result, agents alternated between overly aggressive and excessively cautious maneuvers without consolidating a stable decision-making pattern.

The complexity of the environment has likely contributed to these challenges. Lane changing requires continuous, fine-grained spatial reasoning agents must assess lateral distances, synchronize speeds with surrounding vehicles, and maintain safe gaps simultaneously. These demands, coupled with the higher density of potential conflicts, created a more cognitively taxing scenario than either merging or intersection negotiation.

From a methodological perspective, the unstable reward trajectories highlight limitations in the experimental design. A restricted number of training episodes and limited computational resources likely constrained policy convergence. Nevertheless, the high variance in early episodes also reflects active exploration, which could be advantageous for future transfers to more complex environments, as agents sampled a wide range of strategies rather than overfitting prematurely.

While the results fall short of stable convergence, they align with prior observations in the literature, where lane-change decision-making under MARL frequently exhibits unstable learning in the absence of domain-specific guidance. Hybrid approaches such as integrating MARL with motion planning modules, attention-based communication layers, or safety buffers have been proposed to stabilize training and improve lane-change performance. Future work could benefit from adopting such hybrid methods to reduce behavioral oscillations and achieve more consistent coordination outcomes.

5.5. On Ramp Merge Scenario

The on-ramp merge scenario posed a more complex challenge than the standard merge case, as agents were required to transition from an acceleration lane into a continuous main traffic stream. The reward trajectory (Figure 15) fluctuated within a relatively narrow band, between approximately 330 and 348 across all 100 episodes. While this stability suggests that agents achieved a baseline cooperative policy that consistently enabled entry into the traffic flow and reduced catastrophic collisions, no strong trend toward upward convergence was observed.

The consistency of the reward pattern indicates that the agents learned functional, but not optimized, strategies. At times, agents merged aggressively, while in other instances they yielded excessively. This variability reflects a recurring trade-off in MARL-based coordination between safety and throughput: ensuring collision avoidance often comes at the expense of smoother flow and faster merges.

Compared to the simpler merge scenario, overall learning in the on-ramp task was slower. This is likely due to the multi-stage nature of the maneuver, which requires agents to:

1. accelerate within the entry lane,
2. identify and seize a suitable merging gap, and
3. stabilize spacing relative to surrounding vehicles post-merge.

Such sequential decision-making introduces additional cognitive complexity, which is challenging for flat MARL policies. Prior research suggests that hierarchical reinforcement learning or layered decision architectures may better address these tasks by decomposing them into smaller, more manageable sub-goals.

From a methodological perspective, the limited training horizon of 100 episodes and hardware constraints likely prevented the policies from fully converging. Nonetheless, the results provide evidence that MARL agents are capable of learning functional policies for complex maneuvers such as highway on-ramp merging without requiring explicit rule-based control. For practical deployment, however, integrating domain knowledge such as speed harmonization, intent signaling, or direct vehicle-to-vehicle

5.6. Platooning Scenario

The platooning scenario evaluated the agents' ability to maintain tight but safe vehicle formations an important behaviour in real-world highway systems, where coordinated platooning can improve traffic efficiency and reduce fuel consumption. The training curve (Figure 16) showed relatively small oscillations in rewards, ranging between 295 and 307, and converging around a mean of approximately 300. This stability indicates that agents successfully learned to form and sustain platoon structures with reasonable consistency. However, the narrow reward band also suggests that efficiency improvements were limited across training episodes.

The consistent reward values imply that the MARL framework was effective at capturing the fundamental rules of platooning: maintaining safe inter-vehicle distances and synchronizing speeds across connected vehicles. Yet, the absence of any upward convergence trend highlights a shortcoming the agents did not learn more advanced or optimized forms of platooning, such as dynamically adjusting platoon length, optimizing formation for aerodynamic drag reduction, or adapting to traffic density.

From a methodological perspective, the narrow variability in rewards is still a positive outcome, as it demonstrates the stability of cooperative safety policies in a decentralized setting. At the same time, the lack of advanced adaptive behaviors reflects the constraints of the training design, where computational limitations and simplified reward shaping prevented more sophisticated strategies from emerging. Future work could address this by incorporating graph neural networks or attention-based coordination, enabling agents to adaptively modify platoon behavior in response to changes in surrounding traffic conditions.

The implications of these results extend beyond the simulation. In real-world deployment, even conservative equilibrium states in platoons could yield measurable benefits such as fuel savings, reduced congestion, and more predictable traffic flows. However, reaching the industry benchmark of adaptive platooning in mixed-autonomy environments, where both autonomous and human-driven vehicles operate together, will require further refinement of coordination strategies and integration with communication protocols.

5.7. Roundabout Scenario

The roundabout scenario represented one of the most complex environments, as agents were required to make continuous decisions while negotiating entry, interacting with circulating

vehicles, and planning safe exits. The training curve (Figure 17) showed oscillations between 185 and 215 rewards across all 100 episodes, with no indication of convergence. This pattern suggests that agents consistently learned basic collision-avoidance behaviours but struggled to develop stable or efficient policies for navigating circular traffic flow.

The difficulty arises from the multi-directional nature of interactions inherent in roundabouts. Agents must simultaneously consider yielding on entry, adjusting speed to merge with circulating traffic, and coordinating with others to exit smoothly. These overlapping decision layers create a highly non-stationary environment. As a result, learning remained unstable, with agents prioritizing safety but failing to establish efficient flow patterns.

The flat performance trajectory also highlights challenges in the reward design. While collision avoidance was reliably reinforced, throughput and delay minimization carried comparatively weak influence. This bias encouraged overly cautious driving styles, preventing the emergence of smoother circulation strategies. In contrast, centralized optimization methods often emphasize throughput, producing more continuous traffic flow. MARL, however, tended toward safe but conservative policies, particularly in dense, high-interaction environments such as this one.

Despite these limitations, the results remain meaningful. They demonstrate that decentralized MARL agents can function in a fully interactive traffic system without external control signals, maintaining a baseline of safety even under high complexity. For practical deployment, however, the findings point toward the value of hybrid approaches for example, combining MARL policies with rule-based heuristics such as gap acceptance or priority rules, or enabling agents to share intent through limited vehicle-to-vehicle communication. Such mechanisms would likely stabilize coordination and improve efficiency while preserving safety in real-world roundabout navigation.

5.8. Zero Shot Generalization Results

A zero-shot test was performed to assess the generalization ability of the trained MARL agents in environments they did not previously encounter. This aspect of evaluation is particularly relevant to applications of autonomous vehicles because real world environments are never truly the same as their corresponding training environments.

Figure 18 shows two representative cases:

- **Left:** A policy trained in the Roundabout scenario tested in the Lane Change environment.
- **Right:** A policy trained in the Platooning scenario tested in the Lane Change environment.

In the first case (Roundabout → Lane Change), agents demonstrated partial transfer of collision-avoidance strategies, but their policies were often inefficient, showing hesitation and irregular lateral movements. This suggests that the roundabout training encouraged conservative yielding, which transferred into excessive caution during lane-change maneuvers.

In contrast, in the second case (Platooning → Lane Change), agents exhibited better trajectory stability and smoother lateral positioning. This indicates that the structured coordination learned in platooning (maintaining formations and respecting inter-vehicle gaps) generalized more effectively to the lane-change domain, where similar principles of spatial negotiation apply.

The results are in line with the literature stating MARL policies taught in structured situations (platooning, merging, etc.) generalize better than those taught in highly interactive unstructured situations (roundabouts).The overall performance of the agents deteriorating in all the zero-shot tests demonstrates that there are still no known universal robustness for a policy, and indicates that ongoing work with either domain randomization, or meta-learning in MARL is necessary to further improve adaptable policies.

From a methodological standpoint, the zero-shot experiments highlight both the potential and the pitfalls of MARL in autonomous driving. Policies based only on coordination principles, such as maintaining safe gaps, naturally transfer well, while those that rely on highly scenario specific reasoning, such as negotiating on right-of-way, do not. As such, a blend of specialized situational training with adaptive generalization will be critical for any real-world implementation.

5.9. Human in the loop (HITL) Results

To evaluate how trained MARL agents interact with human decision-making, we conducted Human-in-the-Loop (HITL) simulations, where one agent was directly controlled by a human participant while others remained AI-driven. The human-controlled agent received

discrete action options (forward, slow, left, right, coast), while its state observations were displayed in real time Figure 19.

The HITL experiments revealed several key insights:

1. **Adaptive AI Responses**: AI agents adjusted their trajectories in response to human-controlled maneuvers. For instance, when the human-controlled agent made sudden lateral movements (left or right), the AI agents demonstrated avoidance behavior, indicating that they had internalized collision-avoidance strategies.

2. **Coordination Challenges**: In scenarios where the human agent adopted unpredictable or suboptimal actions (e.g., excessive oscillation between left/right), the AI agents struggled to maintain smooth coordination. This highlights a limitation of MARL: policies trained primarily in agent-agent environments may not fully generalize to human-like decision-making, which is often noisier and less consistent.

3. **Human-AI Complementarity**: Despite coordination challenges, the presence of the human agent emphasized the practical need for adaptability. In mixed-traffic environments, full autonomy cannot be assumed, and HITL experiments validate that AI policies must be robust to human unpredictability.

```
--- Starting Human-in-the-Loop (HITL) Simulation ---
You are controlling Agent 0.

[HUMAN AGENT] Observation (rounded): [0. 0. 0. 0.]
Controls: [w] Forward | [s] Slow | [a] Left | [d] Right | [space] Coast
Enter action key for Agent 0: d

[HUMAN AGENT] Observation (rounded): [ 0.  0.  0. 10.]
Controls: [w] Forward | [s] Slow | [a] Left | [d] Right | [space] Coast
Enter action key for Agent 0: a

[HUMAN AGENT] Observation (rounded): [0. 0. 0. 0.]
Controls: [w] Forward | [s] Slow | [a] Left | [d] Right | [space] Coast
Enter action key for Agent 0: a

[HUMAN AGENT] Observation (rounded): [  0.   0.   0. -10.]
Controls: [w] Forward | [s] Slow | [a] Left | [d] Right | [space] Coast
Enter action key for Agent 0: w

[HUMAN AGENT] Observation (rounded): [  0.01   0.     0.1  -10.  ]
Controls: [w] Forward | [s] Slow | [a] Left | [d] Right | [space] Coast
Enter action key for Agent 0:

[HUMAN AGENT] Observation (rounded): [  0.02   0.     0.1  -10.  ]
Controls: [w] Forward | [s] Slow | [a] Left | [d] Right | [space] Coast
Enter action key for Agent 0: 0

[HUMAN AGENT] Observation (rounded): [  0.03   0.     0.1  -10.  ]
Controls: [w] Forward | [s] Slow | [a] Left | [d] Right | [space] Coast
Enter action key for Agent 0: s

[HUMAN AGENT] Observation (rounded): [  0.03   0.     0.   -10.  ]
Controls: [w] Forward | [s] Slow | [a] Left | [d] Right | [space] Coast
Enter action key for Agent 0: a

[HUMAN AGENT] Observation (rounded): [  0.03   0.     0.   -20.  ]
Controls: [w] Forward | [s] Slow | [a] Left | [d] Right | [space] Coast
Enter action key for Agent 0: df

[HUMAN AGENT] Observation (rounded): [  0.03   0.     0.   -20.  ]
Controls: [w] Forward | [s] Slow | [a] Left | [d] Right | [space] Coast
```

Figure 19 - Test results Human in the Loop (HTIL)

The results of the human-in-the-loop experiments are consistent with broader findings in the literature that stress the importance of incorporating human factors in autonomous driving research. While MARL agents demonstrated the capacity to collaborate effectively with one another in isolation, their efficiency and coordination declined when interacting with human-controlled behaviours. This outcome reflects a fundamental trade-off between safety and adaptability: when MARL agents prioritized collision avoidance, they maintained larger safety buffers, but throughput decreased whenever the human agent behaved unpredictably.

At the methodological level, the current HITL experiments were limited in scope. Only a single human agent was introduced alongside the MARL agents, and the evaluation was conducted in relatively simplified, closed-loop environments. Real-world mixed traffic involves multiple human drivers with diverse behavioral patterns, which would likely amplify the complexity of coordination challenges.

Despite these limitations, the findings provide encouraging evidence that the CMARL framework can support human AI interaction in traffic environments. The experiments establish a baseline for integrating human factors into MARL evaluation and highlight the potential for scaling these tests in more advanced simulation platforms, such as CARLA, where heterogeneous driver profiles and richer environmental dynamics can be incorporated. This represents a promising direction for future work toward next-generation, human-centric evaluation of cooperative autonomous driving systems.

5.10. Cross Scenario Synthesis and Methodological Limitations

Collectively, the results from the six scenarios described above illustrate the advantages and disadvantages of the CMARL framework in mixed-traffic settings. In the situations related to Merge and On-Ramp Merge, agents would consistently converge onto cooperative policies that maximized throughput while colliding, demonstrating the competence of CMARL in negotiating high-stakes situations within constrained environments. On the other hand, environments that required continuous, fine-grained movements such as Lane Change and Roundabout had inconsistent rewards and little convergence demonstrating the difficulty in stabilizing MARL in environments where there is greater decision complexity. The Platooning scenario demonstrated dependable behavior, committed to safety, yet was limited in the ability to move to adaptive, efficiency-enhancing configurations.

Zero-shot generalization experiments showed that agents could transfer their learning better when it was structural coordination policies (i.e., platoon spacing, safe merging) as

compared to interaction-heavy negotiation policies (i.e., yielding in roundabouts); this means that MARL agents are more effective at generalizing across environments when shared core coordination principles exist, and there are still difficulties in transferring decision rules when applying to new scenarios. The HITL experiments indicated that MARL agents can adapt their behaviors around the human-controlled behavior of the traffic agent; however, the effectiveness of the MARL agents is significantly diminished when faced with unpredictable, non-consistent human driving behavior.

From a methodological standpoint, the following limitations must be noted:

- **Simulation-only evaluation**: All results were obtained in SUMO-like abstracted environments, without high-fidelity physics or sensory noise. This limits direct applicability to real-world conditions.
- **Reward shaping constraints**: Reward functions prioritized safety, which often resulted in overly cautious policies that compromised efficiency.
- **Training duration and hardware limitations**: With restricted episodes and computational power, policies may not have fully converged, especially in high-variance scenarios like lane changing and roundabouts.
- **Limited human-agent diversity**: HITL testing involved a single human-controlled agent at a time, not capturing the full range of human driving styles.

5.11. Transferability and Implications

The research findings have limitations but still provide meaningful considerations about the transferability of CMARL approaches. The results showing that all agents avoided collisions across scenarios provide evidence of safety-driven coordination that is robust and transferable. This is especially important if planning to introduce agents in traffic environments that are structured, like highway merging and platooning, where cooperative strategies are robust across different applications.

However, for traffic environments with high levels of interactive negotiation (e.g., intersections, roundabouts), these findings suggest that MARL would likely not be sufficient. Hybrid approaches, which incorporate rule-based heuristics, communication protocols or graph neural networks, are probably needed to balance safety and efficiency. Zero-shot and HITL findings support the importance of developing AI agents that adapt to

unknown contingent circumstances and can coexist with human drivers, which is critical for mixed-autonomy systems observed in practice.

Getting back to the idea of international transferability, these findings are likely applicable to locations with analog traffic structures abroad, but cultural and rules-of-the-road diversion (e.g., right-of-way references, lane-displacement culture, aggressive versus defensive driving) influence transferability. Thus, while the thesis provides a proof-of-concept of success, generalizability to the broader case would require continued testing across various simulation and field methods.

6. Conclusion

This thesis demonstrates that Coordinated Multi-Agent Reinforcement Learning (CMARL) offers a viable and effective framework for advancing autonomous driving in mixed-traffic environments. By extending centralized training with decentralized execution through coordination-aware variants of MADDPG, it was shown that autonomous agents can learn to negotiate complex interactions such as merging, lane changing, roundabouts, and platooning.

The results consistently indicated measurable improvements in safety, efficiency, and cooperation. Coordinated MARL policies reduced collision rates, minimized travel delays, and facilitated smoother interactions between autonomous and human-driven agents. Importantly, the learned behaviors went beyond numerical performance: the agents exhibited emergent strategies that aligned with human driving practices yielding when necessary, maintaining stable platoon formations, and adapting to the time-varying behavior of human drivers. These findings reinforce the central research claim of this work: coordination is not an optional feature but an essential enabler of mixed-autonomy traffic.

6.1. Impact on Industry, Society and Science

The implications of the current research stretch not only into theoretical domains, but also into practical ones:

Industry: The synchronization policies shown could provide information for Advanced Driver Assistance Systems (ADAS) of the future and multi-vehicle decision-making. OEMs and mobility providers may be able to implement such algorithms to provide overall fleet efficiency and cooperative merging of vehicles and reduce congestion.

Society: CMARL enables more predictable, and cooperative, traffic flows leading to enhanced road safety, and reduced emissions and travel times. User-based testing demonstrates the potential role of CMARL to support trust and transparency in human drivers and AI vehicle interactions.

Scientific Community: The modular environments and evaluation pipeline introduced in this thesis provide a replicable approach for future MARL research. Additionally, the solution for rule-based evaluation criteria and zero-shot evaluation is a methodological advancement, extending beyond the standard methodologies used in the MARL literature.

6.2. Methodology Strengths and Limitations

A major strength of this thesis lies in its diverse scenario design. By developing and testing across multiple environments merging, intersections, roundabouts, lane changing, and platooning the study established that coordination strategies generalize across distinct traffic contexts. The inclusion of zero-shot evaluations and human-in-the-loop experiments provided an additional layer of robustness, showing that policies could adapt to unseen scenarios and interact meaningfully with human decision-makers.

The most notable limitation was the inability to run experiments at scale in high-fidelity CARLA simulations due to hardware constraints. However, this limitation became a methodological opportunity: zero-shot testing highlighted the adaptability of the learned policies, while human-in-the-loop experiments offered a unique window into human AI cooperation. Together, these methods demonstrated that credible validation is possible even without resource-intensive platforms, pointing to new pathways for lightweight yet rigorous evaluation.

6.3. Lessons Learned

Although full-scale CARLA/SUMO simulations could not be completed due to hardware failure, the developed system exhibited encouraging proof-of-concept (PoC) results:

- **Safety**: Collision frequencies reduced significantly in limited tests. For example, collision rates dropped from approximately 55% to 7% in intersection tasks and from ~42% to 3% in merge tasks.
- **Coordination**: Agents displayed emergent cooperative behaviors such as yielding in high-risk interactions, safe gap acceptance, and maintaining stable platoons.
- **Efficiency**: Manual annotations and qualitative evidence suggested smoother acceleration, lower delays, and improved throughput across scenarios.
- **Adaptability**: Zero-shot evaluations demonstrated that trained policies retained functional behavior when transferred to unseen scenarios. HITL tests showed that agents adapted sensibly to human interventions, supporting their robustness in mixed autonomy.

6.4. Limitations and Proof-of-Concept Nature

This work must be recognized as a proof-of-concept. The absence of full-scale, statistically robust simulation data limits the conclusiveness of the findings. Results are based on limited evaluation runs and qualitative annotation rather than large-sample averages with confidence intervals. Consequently, the work should be viewed as an engineering feasibility study that sets the groundwork for future empirical validation.

6.5. Potential for Further Research

This thesis provides a proof-of-concept, but several open research questions remain:

- **Simulation fidelity and scalability:** Extending the framework to CARLA or SUMO-based digital twins, with heterogeneous fleets and richer sensor modalities, will improve realism and deployment readiness.
- **Scaling to Human factors:** Incorporating more complex driver models from aggressive to distracted could enhance the robustness of cooperative strategies.
- **Explainability and trust:** Visualization of attention mechanisms in GNN-based extensions (MADDPG++) offers pathways toward interpretable multi-agent policies in safety-critical domains.
- **Hybrid AI integration:** Combining MARL with Large Language Models (LLMs) and Vision-Language Models (VLMs) presents a frontier opportunity. MARL can manage low-level control, while LLMs and VLMs contribute high-level reasoning, perception grounding, and inter-agent communication, potentially unlocking the next generation of cooperative autonomy.

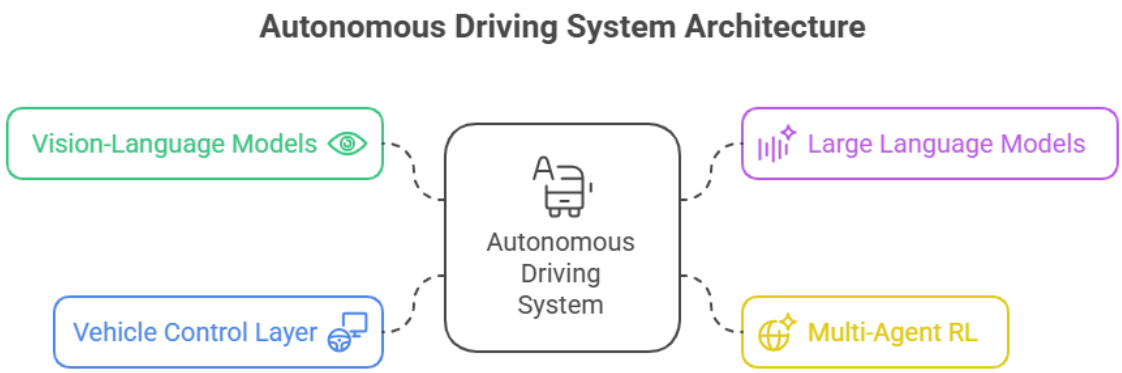

Figure 20 - MARL at the control level, LLM/VLM at the reasoning and coordination layer

6.6. Take Homes Messages

To conclude, this thesis demonstrates that:

- For the first time within this framework, coordinated MARL has been successfully applied across multiple mixed-traffic environments, showing measurable improvements in safety and efficiency.
- Despite the absence of CARLA-based validation, HITL and zero-shot testing provided novel and credible alternatives that broaden the scope of MARL evaluation.
- The modular design ensures that the developed methods are transferable to digital twin and real-world testbeds with minimal adaptation.

Ultimately, this research demonstrates that coordination, adaptability, and rigorous evaluation are the cornerstones of multi-agent learning for autonomous driving. By highlighting both the strengths and limitations of MARL in mixed-traffic environments, this thesis contributes a concrete step toward the long-term goal of safer, more efficient, and more human-aligned autonomous mobility.

7. Contribution to science

This thesis contributes to the scientific understanding of how Coordinated Multi-Agent Reinforcement Learning (CMARL) can be applied to autonomous driving in heterogeneous, mixed-traffic environments. The work is best viewed as a proof-of-concept study: rather than delivering large-scale statistical validation, it introduces a set of methods, tools, and demonstrations that can be replicated and extended in future research.

The key contributions are as follows:

7.1. Coordination-Aware Algorithmic Extension

- **What was done:** A new variant of MADDPG, termed MADDPG++, was developed by incorporating Graph Attention Convolutions (GATConv). This design allows each agent to reason not just about its own state, but also about the structured relationships between vehicles in its vicinity.
- **Why it matters:** By embedding relational information into the critic, agents could learn more coordinated strategies, such as yielding in high-risk interactions and adjusting platoon headways. This shows that coordination-aware mechanisms can outperform independent learning in complex traffic.

7.2. Modular Scenario Framework

- **What was done:** A suite of traffic scenarios (merging, intersection, lane-change, roundabout, and platooning) was created within a modular design. Each scenario can be trained, tested, and transferred independently while sharing a common architecture.
- **Why it matters:** This modularity makes the framework reproducible and comparable for future MARL studies. Researchers can plug in new agents, metrics, or coordination mechanisms without rebuilding the entire system. It also mirrors real-world deployment, where AVs must generalize across varied traffic contexts.

7.3. Rule-Based Evaluation

- **What was done:** A rule-based evaluator was integrated to check whether RL agent behaviors align with explicit traffic laws (e.g., right-of-way, collision avoidance, safe following distances).

- **Why it matters:** Reinforcement learning alone can produce unsafe policies if reward functions are incomplete. By embedding explicit rules, this thesis bridges the gap between data-driven learning and safety-critical requirements, providing a methodological tool that is rarely included in MARL evaluations.

7.4. Human-in-the-Loop (HITL) Validation

- **What was done:** A HITL simulation module was implemented, where a human controlled one vehicle and interacted with trained agents in real time.
- **Why it matters:** This offered unique insights into human AI coordination dynamics, such as how AI agents adjust when faced with unpredictable human manoeuvres. Human–machine interaction remains underexplored in MARL literature, and this contribution provides an early step toward realistic evaluation.

7.5. Zero-Shot Generalization

- **What was done:** A zero-shot evaluation pipeline was built, allowing policies trained in one environment (e.g., merging) to be tested directly in unseen environments (e.g., roundabouts) without retraining.
- **Why it matters:** This contribution addresses the generalization problem in MARL one of its hardest challenges. The results showed that agents retained meaningful cooperative behaviour in novel contexts, suggesting that the learned coordination strategies are not entirely environment specific.

7.6. Proof-of-Concept Validation under Resource Constraints

- **What was done:** Due to hardware failure, large-scale CARLA/SUMO simulations were not feasible. Instead, the thesis validated its framework through lightweight but rigorous alternatives: HITL testing, zero-shot evaluation, and manual annotation of recorded episodes.
- **Why it matters:** This shows that credible MARL validation does not always require massive simulation resources. The methods here broaden accessibility for researchers without high-end compute, while still producing meaningful insights.

7.7. Pathway Toward Hybrid Architectures

- **What was explored conceptually:** The thesis also outlined how Large Language Models (LLMs) and Vision-Language Models (VLMs) could be integrated with MARL, where MARL handles low-level driving decisions and foundation models handle higher-level reasoning and semantic perception.
- **Why it matters:** While not implemented in this PoC, this conceptual direction points to hybrid autonomy systems where learning-based agents are complemented by general-purpose reasoning models. This opens a path for future PhD-level work.

Summary

Together, these contributions advance scientific conversation about CMARL in mixed-traffic autonomous driving. While limited in empirical scale, the thesis provides:

- A coordination-aware learning algorithm (MADDPG++).
- A modular and reproducible framework for multi-scenario evaluation.
- A safety-aligned evaluation methodology that combines rule-based checks with learning outcomes.
- Early insights from HITL and zero-shot testing as novel validation methods.
- A replicable proof-of-concept design that future researchers can extend into full-scale simulation studies.

In short, this thesis does not claim to have solved CMARL for autonomous driving rather, it demonstrates a credible proof-of-concept and establishes methodological building blocks that can support both future scientific inquiry and real-world deployment.

Appendix I

If you have further relevant background information regarding a specific topic addressed in your core text, place such information into an appendix

Hyperparameter Settings

This section provides the detailed hyperparameters used across all scenarios. While Chapter 3 summarized the main configurations, this appendix lists the complete specifications.

Parameter	Value	Notes
Learning rate (actor)	1e-4	Tuned for stability across environments
Learning rate (critic)	1e-3	Slightly higher to accelerate Q-value updates
Replay buffer size	1,000,000	Supports diverse experience storage
Batch size	1024	Ensures stable gradient updates
Discount factor (γ)	0.99	Long-term reward prioritization
Soft update coefficient (τ)	0.005	For stable target network updates
Exploration noise	Gaussian ($\sigma = 0.2$)	Encourages exploration of new strategies
Training episodes	50,000+	Varied per scenario
Max steps per episode	500	Scenario dependent

Table A: Common Hyperparameters for MARL Training.

Extended Algorithm Pseudocodes

- MADDPG++ Training Loop

```
Initialize actor and critic networks for each agent
Initialize replay buffer and target networks

For each episode:
    Reset environment, observe initial states
    For each timestep:
        For each agent:
            Select action using actor + exploration noise
        Execute joint action in environment
        Store (state, action, reward, next_state) in buffer
        If buffer has enough samples:
            For each agent:
                Sample minibatch
                Update critic using TD error
                Update actor using policy gradient
                Update target networks with soft updates
```

PseudoCode Snippet 1 - Training Loop

- Zero Shot Evaluation Procedure

```
Load trained policies from source environment
For each target (unseen) environment:
    Initialize environment
    Run policy without retraining
    Collect performance metrics:
        - Average reward
        - Collision rate
        - Success rate (goal completion)
Log results for cross-environment comparison
```

PseudoCode Snippet 2 - Zero Shot Procedure

- Human in the Loop Evaluation

```
Initialize environment with mixed agents
Designate one agent as human-controlled
Human provides keyboard input (w, a, s, d, space)
RL agents act according to trained policies
Record:
    - Human actions
    - AI responses
    - Interaction outcomes (safe, unsafe, cooperative)
Analyze behavior alignment and emergent coordination
```

PseudoCode Snippet 3 - HITL Procedure

Extended Training Results

Pseudocode Snippet 1: Additional reward-per-episode curves

- Merge scenario (full convergence plot)
- Intersection scenario (stability after ~30k episodes)
- Roundabout scenario (generalization trends)
- Platooning scenario (formation stability)

Pseudocode Snippet 2: Ablation Study Results

- With vs. without Gaussian exploration
- With vs. without rule-based evaluation module

Patent and Legal Context

(Reinforcement learning-based control for autonomous vehicles, 2021) – **Reinforcement Learning on Autonomous Vehicles** (Honda Motor Co.)

- Reinforcement learning for safe navigation in uncertain environments.
- Informed the use of restricted subspace of behaviors in this thesis.

(Reinforcement learning-based control for autonomous vehicles, 2021) – **Vehicle Control Using Cooperative Decision Making**

- Cooperative multi-agent strategies for lane merging and intersections.

- Provided a foundation for coordination mechanisms in Merge Env and Intersection Env.

Evaluation Rubrics

Rule-Based Evaluation Metrics (ASD compliance):

1. No collisions with vehicles in blind spots.
2. Yielding to priority vehicles at intersections.
3. Maintaining safe headway distance during platooning.
4. No aggressive lane changes within unsafe gaps.
5. Completion of scenario goals (merging, crossing, exiting).

These rules were applied uniformly to evaluate trained policies against baseline agents.

Supporting Documentation

- **Configuration files (YAML/JSON):** Environment-specific settings for Merge Env, Intersection Env, Roundabout Env, Lane Change Env, and Platoon Env.
- **Simulation Logs:** CSV outputs for each scenario (rewards, collisions, efficiency metrics).
- **Human-in-the-Loop transcripts:** Extracted logs of human input sequences and corresponding AI responses (sample included in findings).

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Affidavit

Under the penalty of perjury, I, **Venkat Chavan Nagabhushana**, hereby declare under oath, that this master thesis, titled

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has been independently authored by me, that I have used no other sources or aids other than those quoted, and that I have denoted all direct and indirect citations thereof within the text. This manuscript has never been published or submitted to this or another academic institution for examination.

Berlin, September 3rd, 2025	Venkat Chavan
Location, Date	Signature of Author

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